

Exact Unlearning from Proxies Induces Closeness Guarantees on Approximate Unlearning

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Abstract

This paper proposes a paradigm shift linking machine unlearning directly to the structure of the data distributions rather than a mere update of the neural network parameters. We show that inferring these distributions with precision enables distilling the exact unlearning signal induced by the modeling. Theoretical bounds on the Kullback-Leibler divergence from the ideal retrained model to our unlearned model, under verifiable admissibility criterion, reveal the soundness of our framework. This method is experimentally validated over three forgetting scenarios as reaching the closest classifier to the ideal retrained model when compared to competitors.

1 Introduction

Approximate machine unlearning removes the influence of some training data from an already trained classifier, without retraining it from scratch. Application ranges from privacy issues, the right to be forgotten (European GDPR, Canadian PIPEDA or Californian CCPA), to security defenses removing poisoned training data. Another scenario is the derivation of a restricted public model from a powerful private model learned on some sensitive data.

Classical methods construct asymmetrical loss that treat retain and forget samples differently. They are iterative, hyper-parameter sensitive, and without stopping criterion. Their performances are often gauged by misleading accuracy metrics and lack of a measurable distance to the retrained model.

This paper first models the exact unlearning signal as the logits shift between the ideal and the initial NNs. Our update is exact up to the quality of the chosen proxies modeling the posterior distributions (see Prop. 1). Section 3 bounds the gap to the ideal retrained model under verifiable assumptions about the proxies. The bound tightens as the proxies quality increases. Implementing this unlearning signal in the NN parameters is only a secondary matter achieved through distillation.

The key element of our method is the choice of the proxies modeling the posterior distributions. Naive marginalisation is a first approximation which turns out to be limited in challenging scenarios. Section 4 introduces some mathematical modeling specifically designed for machine unlearning.

Our contributions are the following:

- An exact unlearning signal, approximately computed thanks to proxies with theoretical guarantees and computable bounds on the gap to the ideal retrained model (Sect. 3).
- A series of fine-grained proxies modeling the posterior distributions that intrinsically learns the membership dependency from the data structure (Sect. 4). Among them, the feature agnostic unlearning DIR-2C naturally emerges from the empirical measure of the data.
- A way to assess unlearning by the empirical evaluation of KL divergences (App. G).
- An experimental validation across four architectures (including a foundation model backbone) and three unlearning scenarios: random data, subclass, or class forgetting. (Sect. 5)

2 Related work

Closed-form Modeling of class-conditional densities has a rich history in statistics. Gaussian discriminant analysis (LDA, QDA) provides the canonical example, with regularized variants [Friedman, 1989] and Bayesian counterparts [Murphy, 2012, Gelman et al., 2013]. More flexible options include per-class Gaussian mixtures [Dempster et al., 1977, McLachlan and Peel, 2000] and constrained covariance structures [Ghahramani and Hinton, 1996]. Our method leverages these fast closed-form approaches. To our knowledge, no prior work has exploited the structure that arises when retain and forget samples within a class are modeled as two separate subclasses.

Machine unlearning is the art of removing the influence of some training data without learning a new model from scratch (see App. A for a deeper overview). The gold standard is *exact removal*. As far as classifiers are concerned, exact techniques resort to specific data structures to avoid full retraining [Cao and Yang, 2015, Sekhari et al., 2021]. For instance, Bourtole et al. [2021] propose the method SISA, which splits the training data into shards and stores intermediate checkpoints for each shard. The trade-off between the memory footprint and the complexity of unlearning is the main limitation at scale, especially when the training data to be forgotten spans multiple shards.

Approximate unlearning aims at making the output of the unlearned model statistically indistinguishable from the ideal model that never saw the forget data. The dominant paradigm, so-called first-order methods, resorts to gradient-based updates of the NN parameters. Gradient ascent on the forget-set loss is the simplest primitive, used in many methods like [Graves et al., 2021, Chen et al., 2023]. Knowledge-distillation variants enforce retention jointly with forgetting. SCRUB trains a student by minimising KL divergence against the original model on the data to retain while maximising the loss on the data to forget [Kurmanji et al., 2023]. Chundawat et al. [2023] use an incompetent teacher guiding the student to suppress class knowledge. Tarun et al. [2023] propose an impair-repair scheme: an error-maximising noise degrades class knowledge, then a repair step restores retain-set accuracy. SalUn restricts updates to the weights most responsible for forgotten data [Fan et al., 2024].

Recent articles unify these first-order methods into either a multi-objective [Zhou et al., 2025, Pan et al., 2026] or a constrained optimization [Dine et al., 2025, Kim et al., 2025]: unlearning the forget data while preserving the model utility. These four papers outline the conflicting gradients of the objectives and propose ad-hoc combinations. Yet, the iterative updates result in a trajectory with no clear stopping criterion. In the end, opacity prevails since there is no characterization of the result in terms of data distribution. Last but not least, Mavrothalassitis et al. [2026] recently spotted that the dependence between the forget and retain data frequently prevents these first-order methods from performing machine unlearning properly.

From the theoretical point of view, KL divergence from the ideal model to the empirically unlearned NN has been adopted as a natural criterion [Sekhari et al., 2021, Golatkar et al., 2020a, Chien et al., 2024, Georgiev et al., 2026]. Yet, nearly all theoretical results are stated in terms of NN properties (Hessian approximation, Lipschitz constants of the loss) rather than of data distribution. Indeed, Triantafillou et al. [2024] report from the NeurIPS unlearning competition that the field still relies on membership inference attacks (MIAs) [Shokri et al., 2017, Carlini et al., 2022, Hayes et al., 2025] as a proxy for formal guarantees, underscoring the lack of interpretable, data-driven bounds.

Our positioning Georgiev et al. [2026] propose data model matching as a way to predict the output of the retrained model but we show that modeling only the retrain set is sub-optimal. Le et al. [2026] give guarantees but primarily designed for the class-wise unlearning scenario. Our work departs from all the above by providing a closed-form decomposition of the expected KL divergence directly in terms of the quality of mathematical models fitted on the data. First, exact unlearning signal is designed as a correction in the logit space from explicit mathematical models. For instance, the LDA-Mix and QDA-Mix constructions picture the retain and forget sub-populations within each class as two Gaussian distributions enabling the removal of order 2 statistic of the true unlearning signal. It yields a precise correction capturing intra-class structure invisible to standard methods. Our proxy LDA-2C goes even further: doubling the label space to intrinsically capture the retain-forget distribution overlap, implicitly stabilizing the covariance on the forget set and extracting an even more precise unlearning signal. The network weight update only emerges as a second distillation step.

3 Our method

3.1 Notations (summarised in Table 4, App B)

The original training set $\mathcal{D} = \mathcal{D}_r \cup \mathcal{D}_f$ is composed by the forget set $\mathcal{D}_f$ to be erased and the retain set $\mathcal{D}_r$ to be preserved. Given a neural network (NN) f_θ trained on $\mathcal{D}$, machine unlearning seeks an efficient procedure producing NN $f_{\tilde{\theta}}$ as if it had been trained on $\mathcal{D}_r$ alone.

Here, NNs are functions from the input space $\mathcal{X}$ to the *logit* space $\mathbb{R}^C$, with C classes. The softmax operator yields predicted class probabilities conditioned on x , a.k.a. *probits*, $p_\theta(x) \doteq \text{softmax}(f_\theta(x))$ belonging to the simplex of probability distributions. Conversely, $f_\theta(x) = \log(p_\theta(x)) + \gamma(x)$ with $\gamma(x) \doteq \text{LSE}(f_\theta(x))$, LSE being the Log-Sum-Exponential function (see App. C.2).

We denote by $\mathbb{P}$ the true joint distribution of the data (Y, X) . The induced marginal and conditioned true distributions are denoted $\mathbb{P}_X$, $\mathbb{P}_Y$ and $\mathbb{P}_{Y|X}$, $\mathbb{P}_{X|Y}$, or simply $\mathbb{P}$ when the context is clear.

We introduce a third representation of the distribution of the data given by a proxy $\mathbb{P}_{X|Y}$, *i.e.* a parametric mathematical model of the true class-conditional distribution $\mathbb{P}_{X|Y}$. This proxy induces the following posterior and associated ‘logit’ vector $M(x) \in \mathbb{R}^C$:

$$\forall y \in [C], \forall x \in \mathcal{X}, \quad \mathbb{P}_{Y|X}(y|x) \doteq \frac{\mathbb{P}_{X|Y}(x|y) \mathbb{P}(y)}{\sum_{c=1}^C \mathbb{P}_{X|Y}(x|c) \mathbb{P}(c)}, \quad M(x)_y \doteq \log \mathbb{P}_{Y|X}(y|x). \quad (1)$$

Again, subscripts may be omitted if the context is clear. Each of these objects has its counterpart denoted with subscript r when related to the retain set $\mathcal{D}_r$. This includes the ideal NN f_{θ_r} learned from $\mathcal{D}_r$, the true distribution $\mathbb{P}_r$, and the proxy $\mathbb{P}_r$.

Kullback-Leibler divergence from distribution $p_{Y|X}$ to $q_{Y|X}$ is denoted as $\text{KL}_{Y|X}(p||q) \doteq \sum_{y=1}^C p_{Y|X}(y) \log(p_{Y|X}(y)/q_{Y|X}(y))$. Its expectation over $\mathbb{P}$ is denoted $\mathbb{E}\text{KL}_{Y|X}(p||q)$.

3.2 A family of unlearned models

For any first-order method mentioned in Sect. 2, the NN update $\Delta(x) = f_{\tilde{\theta}}(x) - f_\theta(x)$ cannot be quantified in terms of unlearning efficacy: it is a function of the optimization trajectory and carries no information about the ideal classifier $\mathbb{P}_{r,Y|x}$. Our key idea is to replace Δ by the proxy shift $\Delta M(x) \doteq M_r(x) - M(x)$, computed from the explicit parametric proxies of the data structure, for which the shift is fully interpretable. To our knowledge, this is the first work to provide a closed form, interpretable decomposition of the unlearning error in terms of end-to-end proxies quality.

We indeed consider a family of unlearned models defined in the logit space, for all $0 \leq \eta \leq 1$:

$$\tilde{f}_\eta(x) \doteq f_\theta(x) + \eta \Delta M(x), \quad \Delta M(x) \doteq M_r(x) - M(x), \quad \forall x \in \mathcal{X}. \quad (2)$$

We find the optimal value $\eta_{\max}$ as the upper bound of a theoretically safe regime (see App. C.1 and C.2). The final proposed unlearned model is then this precise $\tilde{f}_{\eta_{\max}}$. The update $\eta_{\max} \Delta M$ may be implemented as a logit processor on top of the initial classifier f_θ taken as a black-box provided it outputs a logit vector. If f_θ is a white box NN, then one may cast this update directly in its parameters. The standard option fine-tunes the NN parametrized by θ with a the teacher-student distillation:

$$\min_{\delta\theta} \mathbb{E}_{\mathcal{D}} \text{KL}_{Y|X}(\tilde{p}_{\eta_{\max}} || p_{\theta+\delta\theta}), \quad \text{with } \tilde{p}_{\eta_{\max}} \doteq \text{softmax}(f_\theta + \eta_{\max} \Delta M). \quad (3)$$

3.3 Assumptions

A1 The data are an independent sampling of the true distributions: $\mathcal{D} \sim \mathbb{P}^{|\mathcal{D}|}$ and $\mathcal{D}_r \sim \mathbb{P}_r^{|\mathcal{D}_r|}$.

A2 The original true distribution is a mixture of form [Le et al., 2026]:

$$\mathbb{P} = \pi_r \mathbb{P}_r + \pi_f \mathbb{P}_f \quad \text{with} \quad \pi_r + \pi_f = 1. \quad (4)$$

A3 The initial NN perfectly captures the true distribution of the data: $\forall x \in \mathcal{X}, p_\theta(x) = \mathbb{P}_{Y|X}(\cdot | x)$.

Definition 1 (Admissible proxies). *We say that an ordered pair of proxy distributions $(\mathbb{P}, \mathbb{P}_r)$ is $\mathbb{P}$ -admissible if, and only if, it verifies the equation:*

$$\mathbb{E}\text{KL}_{Y|X}(\mathbb{P} || \mathbb{P}) \leq \mathbb{E}\text{KL}_{Y|X}(\mathbb{P} || \mathbb{P}_r). \quad (5)$$

A4 $\mathbb{P}$ is a better proxy for $\mathbb{P}$ than $\mathbb{P}_r$, in the sense that $(\mathbb{P}, \mathbb{P}_r)$ is $\mathbb{P}$ -admissible.

Remarks. Assumption A3 is the objective of cross-entropy training: minimizing it drives p_θ to $\mathbb{P}_{Y|X}$ (App. E, Fig. 3). It enables splitting the learning from the unlearning errors (App. D bounds the cost of relaxing it by $\mathcal{O}(\epsilon)$). Assumption A4 is verifiable experimentally before starting the unlearning process. Our experiments in Sect. 5.2 show that the better the proxy, the more likely it will verify this assumption.

3.4 Theoretical guarantees

This section gives a sketch of the reasoning, which is fully detailed in App. C and D.

First, App. C.1 justifies our choice (2) of unlearned models. These are indeed a family of solutions to an optimization problem of information geometry, looking for an update similar to the initial NN p_θ , but closer to the retain proxy P_r than the initial proxy P . An interpretation is that an observer seeing samples drawn from the unlearned model $\tilde{p}_\eta$ is likely attributing them to P_r rather than P .

Second, App. C.2 seeks a safety range for η insuring that the unlearned $\tilde{p}_\eta$ provides an improvement. The quantity $\eta_{\max}$ denotes the biggest scale at which one can amplify the exact unlearning signal extracted through the proxies while transferring it into the NN. This justifies our further choice to take $\eta = \eta_{\max}$ in the experiments of Sect. 5. Furthermore, finding $\eta_{\max}$ is simple with a 1-d line search over the empirical evaluation of a convex function.

The main quantity of the study, $\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\tilde{p}_\eta)$, gauges how close the probits $\tilde{p}_\eta$ of the unlearned model are from the ideal distribution given by the conditional true distribution $\mathbb{P}_{r,Y|x}$. We look for a proof of soundness showing that the unlearned model is closer to this ideal than the initial NN.

App. D.2 first expands the divergence to exhibit an interpretation in terms of modeling shift:

Proposition 1. *For any $\eta \in [0, \eta_{\max}]$, the divergence from the ideal retrained model to our unlearning target $\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\tilde{p}_\eta)$ is tightly bounded by:*

$$(1 - \eta) \underbrace{\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P})}_{\text{initial divergence}} + \eta \underbrace{\left(\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P}_r) - \mathbb{EKL}_{Y|X}(\mathbb{P} \|\mathbb{P}) \right)}_{\text{modeling shift}} + \mathcal{O}(\pi_f). \quad (6)$$

The first term of the bound is close to zero either when the posteriors of $\mathbb{P}_r$ and $\mathbb{P}$ are similar (nothing to unlearn), or when η is close to 1. The second term of the bound can be small even if the proxies (P_r, P) may not accurately model their respective targets $(\mathbb{P}_r, \mathbb{P})$. It outlines that we should model both $\mathbb{P}$ and $\mathbb{P}_r$ in a similar way so that their KL divergences compensate. Modeling only the retrain set Georgiev et al. [2026] is sub-optimal. As a final remark, (6) is our tightest bound. It indeed becomes an equality for $\eta = \eta_{\max}$ (see App. D.2).

App. D.3 states another bound which is less tight than (6) but it reflects the intuition of why modeling the conditional distributions $X|Y$ proxies as accurate as possible is an advantage.

Proposition 2. *For any $\eta \in [0, \eta_{\max}]$ we have the intuitive bound:*

$$\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\tilde{p}_\eta) \leq \eta (\mathbb{EKL}_{X|Y}(\mathbb{P}_r \|\mathbb{P}_r) + \mathbb{EKL}_{X|Y}(\mathbb{P} \|\mathbb{P})) + \mathcal{O}(\pi_f) + \mathcal{O}(1 - \eta). \quad (7)$$

Finally, the appendix shows that $\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\tilde{p}_\eta)$ can be exactly written as:

$$\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P}_\theta) + \underbrace{\mathbb{E}[\Delta\gamma(X, \eta)]}_{<0, \forall \eta \in [0, \eta_{\max}]} + \eta \underbrace{\left(\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P}_r) - \mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P}) \right)}_{<0 \text{ if } (P_r, P) \text{ is } \mathbb{P}_r\text{-admissible}}$$

App. D.1 shows that the second term is negative in the safety range. The third term is negative if (P_r, P) is $\mathbb{P}_r$ -admissible, i.e. P_r is a better proxy for $\mathbb{P}_r$ than P . The proof of soundness reads as:

Proposition 3 (Provable decrease of KL). *For any $\eta \in [0, \eta_{\max}]$, if one further assumes that the pair of proxies (P_r, P) is $\mathbb{P}_r$ -admissible,*

$$\mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\tilde{p}_\eta) \leq \mathbb{EKL}_{Y|X}(\mathbb{P}_r \|\mathbb{P}_\theta) \quad (8)$$

The above result is only a sufficient condition over the decreasing. However it suggests that by striving to achieve the best possible modeling of the data and create (P_r, P) an $\mathbb{P}_r$ -admissible pair of proxy, it can only improve the resulting unlearning.

4 Unlearning aware data modeling

This section shows how to take into account the unlearning scenario to construct proxies (P, P_r) that are close to their respective target distributions. This is important for two reasons: i) Assumption A4 is valid, ii) the bound of Prop. 2 is small. Our method can be applied to any type of parametric distributions. This section illustrates it with Gaussian distributions to model $X|y$ (Eq. (1) recovers the posteriors $P_r(\cdot|x)$ and $P(\cdot|x)$), or empirical Dirac distribution to model (X, Y) .

4.1 Naive models

LDA is a first model considering one distribution per class sharing a single covariance matrix, both for P and P_r . There are $2C$ expectation vectors and 1 covariance matrix.

$$P : X|y \sim \mathcal{N}(\mu_y; \Sigma), \quad P_r : X|y \sim \mathcal{N}(\mu_{r,y}; \Sigma). \quad (9)$$

QDA is the variant using one covariance per class. There are $2C$ expectation vectors and C covariance matrices:

$$P : X|y \sim \mathcal{N}(\mu_y; \Sigma_y), \quad P_r : X|y \sim \mathcal{N}(\mu_{r,y}; \Sigma_y). \quad (10)$$

DIR is an alternative based on the empirical distribution, *i.e.* a sum of Diracs

$$P : (X, Y) \sim \mathcal{D}, \quad P_r : (X, Y) \sim \mathcal{D}_r. \quad (11)$$

App. F shows that this creates an unlearned model, independent of $\eta > 0$, s.t. $\tilde{p}(x) = p_\theta(x)$ if $x \notin \mathcal{D}_f$, and for the forget set:

$$\tilde{p}(x)_c = \begin{cases} 0 & c = y(x), \\ p_\theta(x)_c / (1 - p_\theta(x)_{y(x)}) & c \neq y(x). \end{cases} \quad (12)$$

This is useless in practice. Yet, the KL distillation drives the unlearned NN *towards* a total annihilation of the forget set. The global effect is a strong decrease of the probit related to $y(x)$ for the forget data. This is interesting for class-wise unlearning but dramatic for the subclass forgetting (see Sect. 5.1).

4.2 Mixture models

Better modeling takes into account the data from the forget set. Following assumption A2, for each class y , we infer the distribution of the forget set $\mathcal{D}_f(y)$ with a data model $P_f(x|y)$. For each class y , the rules of total probabilities gives:

$$P(x|y) = (1 - \pi_f(y))P_r(x|y) + \pi_f(y)P_f(x|y), \quad (13)$$

where $\pi_f(y) = |\mathcal{D}_f(y)|/|\mathcal{D}(y)|$ is the true intra-class proportion of the forget set. This proxy is more informative yielding an update $\Delta M(x)$ if and only if there exists a natural spread between the forget and the retain set at the raw data level. It precisely checks if there is something to unlearn (or not).

LDA-Mix considers one distribution per class and per state $s \in \{r, f\}$ with a shared covariance per state. There are $2C$ expectation vectors and 2 covariance matrices:

$$P_r : X|y \sim \mathcal{N}(\mu_{r,y}; \Sigma_r), \quad P_f : X|y \sim \mathcal{N}(\mu_{f,y}; \Sigma_f). \quad (14)$$

QDA-Mix is the variant with one covariance per class and state. There are $2C$ expectation vectors and $2C$ covariance matrices:

$$P_r : X|y \sim \mathcal{N}(\mu_{r,y}; \Sigma_{r,y}), \quad P_f : X|y \sim \mathcal{N}(\mu_{f,y}; \Sigma_{f,y}). \quad (15)$$

The method DIR stays unaffected by this awareness.

4.3 Doubling labels

Another refinement artificially considers the variable $(y, s) \in \llbracket C \rrbracket \times \{r, f\}$ as a label itself and produces a mixture on these $2C$ labels. Doing so, the produced proxy on the initial data intrinsically estimate the quantity $P(s|x)$ which helps the unlearning. To our knowledge, no other unlearning work lift the labeled data (x, y) in upper dimension while doubling the label dimension in order to have a richer mixture. This subtle shift results in a final mixture on the posterior rather than on $X|y$. App. F details the projection back to dimension C to update the initial model.

LDA-2C applies a unique LDA on $(y, s) \in \llbracket C \rrbracket \times \{r, f\}$. There are thus $2C$ expectation vectors and 1 covariance matrix, whose estimation is more robust than for LDA-Mix.

$$P : X|y, s \sim \mathcal{N}(\mu_{s,y}; \Sigma), \quad P_r : Y|x \sim P(Y|x, r). \quad (16)$$

QDA remains unaffected by double labels.

DIR-2C

$$P : (X, Y, s) \sim \mathcal{D} \times \{r, f\}, \quad P_r : Y|x \sim P(Y|x, r). \quad (17)$$

App. F shows that this creates an unlearned model, independent of $\eta > 0$, s.t. $\tilde{p}(x) = p_\theta(x)$ if $x \notin \mathcal{D}_f$, and for the forget set:

$$p_{\tilde{\theta}}(x)_c = \frac{|\mathcal{D}_r(y(x))|}{|\mathcal{D}(y(x))|} p_\theta(x)_c + \frac{|\mathcal{D}_f(y(x))|}{|\mathcal{D}(y(x))|} \begin{cases} 0 & c = y(x), \\ p_\theta(x)_c / (1 - p_\theta(x)_{y(x)}) & c \neq y(x). \end{cases} \quad (18)$$

This avoids the total canceling in (12). Again, this unlearned model is useless in practice, but it plays the important role of the teacher in the KL distillation. The effect is a slight decrease of the probit related to $y(x)$ for the forget data.

5 Experiments

We evaluate our unlearning protocol along three axes: (i) it fits a well-defined unlearning signal leading to substantial improvements (Sect. 5.2), (ii) which proxy fits which forgetting scenario or which architecture (Sect. 5.2), and (iii) how the resulting unlearned models compare to the standard SOTA baselines, and the ideal NN trained from scratch (Sect. 5.3). This section reports only a representative subset of our experimental results. A non-exhaustive ablation study is provided in App. I.2. The full set of results is released as a supplementary zip archive, documented in App. I.2.

5.1 Setup

Datasets and architectures. CIFAR-10 ($C = 10$, 2 superclasses) [Krizhevsky, 2009] is the primary testbed. CIFAR-100 ($C = 100$, 20 superclasses) [Golatkhar et al., 2020a] probes scaling to a denser label space. We consider 4 architectures. The first three use **DINOv2-Small** [Oquab et al., 2024] as a frozen backbone foundation model, completed with the heads: **linear** ($384 \rightarrow C$), **mlp1** ($384 \rightarrow 256 \rightarrow C$), or **mlp2** ($384 \rightarrow 256 \rightarrow 256 \rightarrow C$). The fourth architecture is a **ResNet-18** where our method is applied to low-dimension projected features (see App. H).

Three forgetting scenarios. We consider the following scenarios. *Random*: with $\mathcal{D}_f$ sampled uniformly from $\mathcal{D}$ Class where $\mathcal{D}_f$ is an entire class. *Subclass* for CIFAR-10 consist in training a binary classifier ($C = 2$, vehicle vs. animal) and forget one subclass per class at a time. The same holds for CIFAR-100 with $C = 20$ classes grouping each five subclasses. This is the most realistic scenario since the forget set is structured within the classes. This scenario, as expected Mavrothalassitis et al. [2026], causes gradient ascent baselines to systematically fail.

Unlearning methods. We used the five models presented in Sect. 4. Note that class covariance matrices in QDA and QDA-Mix are approximated through their diagonal to obtain a fast inversion process. Six gradient-based baselines covering the standard taxonomy: **FT** (retain fine-tuning), **GA** [Graves et al., 2021], **GA+FT** (one ascent followed by fine-tuning), **RL+FT** [Chen et al., 2023] (random labelling the forget set), **SCRUB** [Kurmanji et al., 2023] (min-max teacher-student), **SalUn** [Fan et al., 2024] (gradient ascent masking and RL+FT). The from-scratch ideal **Retrain** and the initial **Base** NNs are reported as references.

Protocol. Tables report **best epoch** by minimum KL_f on the training data. This selection rule is uniform across methods and maximally favours the SOTA baselines since it lets them back off before destroying the NN’s behavior like SCRUB. Hence Fig. 2 sheds some light on the reading of the benchmark table 3. All presented results are means over 5 seeds, with standard deviation bands in Fig. 2.

Metrics. (i) $\text{Acc}_f, \text{Acc}_t$ are accuracies on forget / full test sets, (ii) KL_f, KL_t are expectations of $\text{KL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}_{\hat{\theta}})$ estimated on the same sets, where $\mathbb{P}_r$ is replaced by the ideal NN f_{θ_r} learned from scratch. KL_{last} reports the KL_t after the last epoch. (iii) In place of an empirical MIA score we report the EER attacker bound $N(\alpha)$ derived in Sect. G (also visualised in Fig. 2, panel (b)). Shadow-model-free empirical MIAs are known to provide unstable signal in the unlearning regime [Hayes et al., 2025]; the Stein bound bypasses this fragility by reading attacker capability directly from KL_f . The raw fast empirical MIA scores were binary with 1 for the initial model and 0 for the retrained and every unlearning methods we tested. Those results remain available in the released results file we provide for inspection. (iv) RTE is the run time efficiency (accounting proxy fit plus distillation for our methods). KL to the retrained model is the primary quality metric: it captures directly the distributional alignment. Acc_f , although widely reported, cannot diagnose this.

5.2 Experimental validation of our method

Convex curves and $\eta_{\max}$. Figure 1 verifies two claims about function $\Delta\gamma$ approximated by h (27): it is convex and admits a unique zero $\eta_{\max}$ in $(0, 1]$. Its value depends a lot on the scenario and the choice of proxies. Of note, $\eta_{\max}$ is not an indicator of an effective unlearning as measured in nats with KL_t or KL_f in Tab. 2. For example, LDA-Mix has small $\eta_{\max}$ values while being a quite effective method. What matters is indeed the scale of the unlearning signal $\eta_{\max}\Delta M$. Prop. 1 backs up this observation showing that the efficacy also depends on the initial divergence and the model shift. Adapting the value of $\eta_{\max}$ is thus of utmost importance.

Table 1 shows that the complexity of the proxies improves the rate at which the assumption A4 is empirically verified. A failure implies that $\eta_{\max}$ is set to 0 and there is no unlearning to perform. Two reasons explain this case. There is nothing to unlearn because $\mathbb{P} = \mathbb{P}_r$. This is often the case with the inconsistent *Random* scenario which randomly forgets few typical data (not outliers). LDA and QDA fail: there is nothing to unlearn in terms of first and second order statistics. The proxy family is too coarse to capture subtle differences between these distributions. The gain of our doubling label over the mixture modeling is seen in the subclass scenario for resnet: LDA-Mix has one fail over the 5 seeds where, then it does not create unlearning signal.

About the distillation. Table 2 shows that, in every case, there is overall no problem in distilling the unlearning signal (3). Surprisingly, the distilled NN $\mathbb{P}_{\hat{\theta}}$ gets slightly better results than $\hat{\mathbb{P}}$. This is due to the fact that the KL divergences are measured from the ideal NN $\mathbb{P}_{\theta_r}$ since $\mathbb{P}_r$ is not available. This gives an advantage to $\mathbb{P}_{\hat{\theta}}$ as it shares the same NN architecture. This advantage is not unfair because getting closer to the ideal NN $\mathbb{P}_{\theta_r}$ is the ultimate goal of machine unlearning.

5.3 Benchmark

This section focuses on *Subclass* scenario because it is more challenging than *Class*, the other scenarios are deferred to App. I.2. The other methods we introduced in Sect. 4 are not part of the following benchmark because we cherry picked the presented here LDA-2C and DIR-2C over the recommendations tables. Moreover we show in Prop. 4 in App. E that the different KL being taken from LDA or QDA proxies are equivalent up to an additional homoscedastic hypothesis cost. The entire database of results is released as a supplementary zip folder detailed in App I.2.

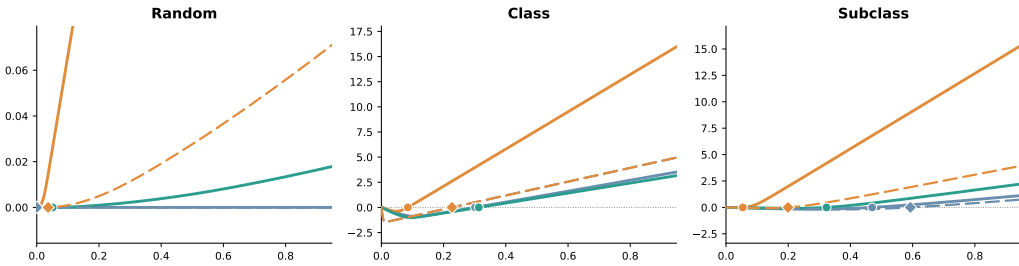


Figure 1: Convexity of $h(\eta)$ on DINOv2+mlp1/CIFAR-10 for 3 forgetting scenarios and 5 proxy models: LDA (9), QDA (10) (dashed), LDA-Mix (14), QDA-Mix (15) (dashed), LDA-2C (16).

Table 1: Recommendations on CIFAR10. Each cell reports (top) $\eta_{\max}$ –mean $\pm$ std over 5 runs– and (bottom) the percentage of runs where A4 holds

	LDA	LDA-Mix	LDA-2C	QDA	QDA-Mix
<i>Across scenarios — arch dinov2-mlp1</i>					
	0.29 $\pm$ 0.00	0.08 $\pm$ 0.00	0.30 $\pm$ 0.00	0.22 $\pm$ 0.00	0.22 $\pm$ 0.00
<i>Class</i>	100%	100%	100%	100%	100%
	0.46 $\pm$ 0.01	0.05 $\pm$ 0.00	0.32 $\pm$ 0.00	0.58 $\pm$ 0.01	0.20 $\pm$ 0.00
<i>Subclass</i>	100%	100%	100%	100%	100%
	0.40 $\pm$ 0.49	0.00 $\pm$ 0.00	0.07 $\pm$ 0.03	0.60 $\pm$ 0.49	0.03 $\pm$ 0.02
<i>Random</i>	40%	80%	100%	60%	100%
<i>Across architectures — scenario Subclass (airplane)</i>					
	0.40 $\pm$ 0.01	0.03 $\pm$ 0.00	0.26 $\pm$ 0.00	0.51 $\pm$ 0.01	0.16 $\pm$ 0.00
Linear	100%	100%	100%	100%	100%
	0.46 $\pm$ 0.01	0.05 $\pm$ 0.00	0.32 $\pm$ 0.00	0.58 $\pm$ 0.01	0.20 $\pm$ 0.00
MLP-1	100%	100%	100%	100%	100%
	0.43 $\pm$ 0.01	0.06 $\pm$ 0.00	0.30 $\pm$ 0.01	0.54 $\pm$ 0.02	0.18 $\pm$ 0.01
MLP-2	100%	100%	100%	100%	100%
	0.43 $\pm$ 0.12	0.02 $\pm$ 0.01	0.32 $\pm$ 0.11	0.24 $\pm$ 0.06	0.09 $\pm$ 0.01
ResNet-18	100%	80%	100%	100%	100%

Table 2: Comparison $\tilde{p}$ vs. $p_{\tilde{\theta}}$ NN on CIFAR10, architecture dinov2-mlp1. Each cell reports KL_t (top) and KL_f (bottom) in nats, mean $\pm$ std over 5 seeds.

	LDA-Mix	LDA-2C	QDA-Mix	DIR	DIR-2C
<i>Class (airplane)</i>					
	0.07 $\pm$ 0.01	0.06 $\pm$ 0.01	0.08 $\pm$ 0.01		0.06 $\pm$ 0.01
$\tilde{p}$	0.72 $\pm$ 0.12	0.63 $\pm$ 0.11	0.80 $\pm$ 0.14	–	0.62 $\pm$ 0.11
	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01
$p_{\tilde{\theta}}$	0.52 $\pm$ 0.08	0.58 $\pm$ 0.09	0.57 $\pm$ 0.08	0.58 $\pm$ 0.09	0.58 $\pm$ 0.11
<i>Subclass (airplane)</i>					
	0.11 $\pm$ 0.00	0.05 $\pm$ 0.00	0.05 $\pm$ 0.00		0.05 $\pm$ 0.00
$\tilde{p}$	1.1 $\pm$ 0.03	0.46 $\pm$ 0.01	0.54 $\pm$ 0.02	–	0.51 $\pm$ 0.00
	0.09 $\pm$ 0.01	0.03 $\pm$ 0.00	0.04 $\pm$ 0.00	0.84 $\pm$ 0.05	0.05 $\pm$ 0.00
$p_{\tilde{\theta}}$	0.87 $\pm$ 0.05	0.29 $\pm$ 0.02	0.42 $\pm$ 0.03	8.8 $\pm$ 0.46	0.48 $\pm$ 0.01
<i>Random (50 samples)</i>					
	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00		0.00 $\pm$ 0.00
$\tilde{p}$	0.15 $\pm$ 0.11	0.09 $\pm$ 0.07	0.13 $\pm$ 0.09	–	0.15 $\pm$ 0.12
	0.03 $\pm$ 0.01	0.03 $\pm$ 0.00	0.02 $\pm$ 0.01	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00
$p_{\tilde{\theta}}$	0.07 $\pm$ 0.08	0.07 $\pm$ 0.05	0.09 $\pm$ 0.08	0.06 $\pm$ 0.04	0.08 $\pm$ 0.07

Criticism of the SOTA baselines. Even if through the lense of accuracies, SOTA methods seems to perform unlearning, Figure 2 shows a more complex reality. SCRUB is not stable in term of KL by design: the gradient ascent to move away from the teacher over the forget set implies a catastrophic behavior. It reaches good statistics in Table 3 because we retain the best KL over the epochs. In practice, without this criterion, its unstable behavior makes it a poor baseline. The other methods reach a ceiling because they quickly over-unlearned. SalUn, GA, RL+FT, and GA+FT destroys the forget accuracy. They cause catastrophic forgetting and without a clear stopping criterion, this instability makes them useless in practice. Fig. 2 shows that FT goes in the right direction but matching the ideal NN p_{θ_r} takes more time than retraining the model from scratch.

Our method spends an overhead to compute the unlearning signal, but the distillation starts producing good NNs from the beginning and quickly converges in Fig. 2. Table 3 shows that distilling from our simplest method DIR-2C already surpasses SOTA baselines in Acc_t , KL_t while on par in RTE. Its main pitfall is that it remains too confident on forget data as its Acc_f is bigger than the accuracy of the ideal NN p_{θ_r} in Table 3. LDA-2C is a richer modeling accounting for the underlying feature distribution. Its distilling outperforms DIR-2C and almost perfectly mimics p_{θ_r} .

Table 3: Benchmark on CIFAR10 / dinov2-mlp1, scenario *SUBCLASS* (airplane). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: best means closest to p_{θ_r} for Acc_f

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.48 $\pm$ 0.01	–	4.8 $\pm$ 0.06	99.7 $\pm$ 0.0	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	95.8 $\pm$ 0.0	61.6 $\pm$ 0.3	100 (ref)
LDA-2C	0.03 $\pm$ 0.00	0.04 $\pm$ 0.00	0.29 $\pm$ 0.02	95.8 $\pm$ 0.1	58.5 $\pm$ 0.7	4.9 $\pm$ 0.07
DIR-2C	0.05 $\pm$ 0.00	0.05 $\pm$ 0.00	0.48 $\pm$ 0.01	99.5 $\pm$ 0.1	98.8 $\pm$ 0.7	4.9 $\pm$ 0.09
SCRUB	0.08 $\pm$ 0.01	4.7 $\pm$ 0.09	0.88 $\pm$ 0.11	95.1 $\pm$ 0.5	55.7 $\pm$ 5.0	6.4 $\pm$ 0.12
SalUn	1.1 $\pm$ 0.05	1.1 $\pm$ 0.05	12 $\pm$ 0.56	89.5 $\pm$ 0.0	0.2 $\pm$ 0.0	4.4 $\pm$ 0.15
RL+FT	0.83 $\pm$ 0.05	0.90 $\pm$ 0.03	8.6 $\pm$ 0.54	89.7 $\pm$ 0.1	2.3 $\pm$ 0.4	4.3 $\pm$ 0.10
GA+FT	1.3 $\pm$ 0.11	1.3 $\pm$ 0.11	14 $\pm$ 1.2	91.2 $\pm$ 0.1	13.7 $\pm$ 0.6	19 $\pm$ 0.30
FT	0.17 $\pm$ 0.02	0.20 $\pm$ 0.04	1.7 $\pm$ 0.21	98.8 $\pm$ 0.2	91.2 $\pm$ 2.1	19 $\pm$ 0.44
GA	9.2 $\pm$ 0.19	858 $\pm$ 33	28 $\pm$ 0.90	60.4 $\pm$ 0.1	0.0 $\pm$ 0.0	2.2 $\pm$ 0.09

6 Conclusion: advantages and limitations

Our theoretical framework introduces $\tilde{p}_\eta = p_\theta + \eta \Delta M$, a closed-form unlearning operator emerging from an explicit constrained optimization problem (App. C). The extracted unlearning signal is derived from any data proxies. We derive bounds (tight in Prop. 1 and pedagogical in Prop. 2) on the KL divergence to the retrained model under a verifiable admissibility criterion A4. The decreasing property in Prop. 3 is a theoretical guarantee that cannot be verified before the unlearning process, contrary to the previous bounds. Three concrete advantages emerge: (i) the procedure unlearns black-box networks outputting logits, (ii) the distilled NNs systematically beats the analytic target itself (Tab. 2), and (iii) the Stein bound (App. G) translates KL_f into a minimum number of attacker queries, replacing fragile empirical MIA scores with an interpretable, closed-form privacy guarantee.

We see the following limitations. (i) The protocol assumes the data controller still holds $\mathcal{D}_r$ to fit P_r . As a countermeasure, DIR-2C only needs $\mathcal{D}_f$ and the proportions π_f label-wise. Also, the distillation may use a data set $\mathcal{D}_{r'} \cup \mathcal{D}_f$ different than $\mathcal{D}$. (ii) In high feature dimension, as for ResNet, estimating LDA/QDA covariances become ill-conditioned. As a countermeasure, we resort to dimension reduction by projecting features onto a small subspace and restrict covariances to diagonal matrices (App. H). (iii) Admissibility A4 is checked on the empirical p_θ before unlearning; admissibility on the true $\mathbb{P}$ therefore holds only up to $\mathcal{O}(\epsilon)$ of good training assumption A3. (iv) The decrease property 3 cannot be verified before unlearning because the unlearner do not have access to $\mathbb{P}_r$ to verify that (P_r, P) is $\mathbb{P}_r$ -admissible. (v) Our analysis proposes a short list of possible proxies without any claim about their optimality.

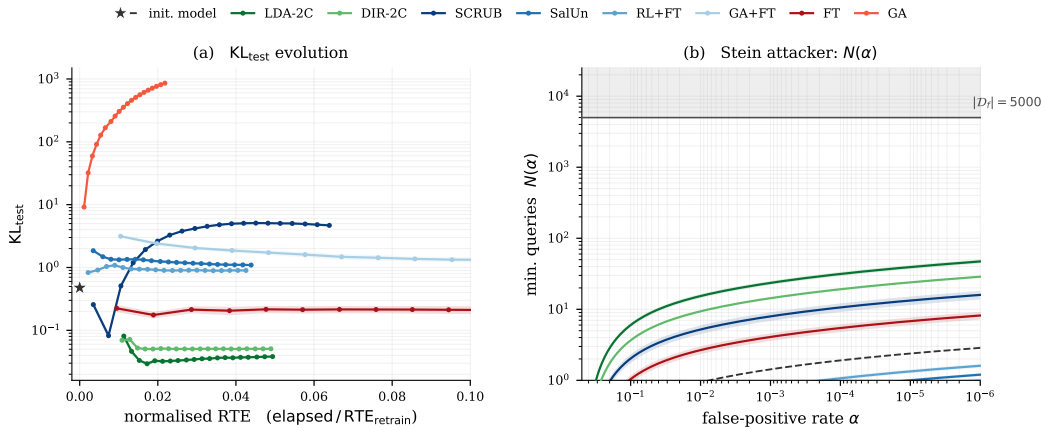


Figure 2: (a) KL_t as a function of RTE normalised by the Retrain RTE on CIFAR-10 / DINOv2 + MLP-1, scenario *Subclass* (airplane). (b) Number of queries an attacker would need to distinguish the unlearned model from the retrained model up to an α rate of false positive.

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A Machine unlearning overview

This appendix expands the related work presented in Sect. 2.

A.1 Closed-form modeling

Closed-form modeling of class-conditional densities has a rich history in statistics. Among these, LDA is the simplest closed-form maximum-likelihood classifier under class-conditional Gaussian data with a shared covariance [McLachlan, 1992] (see App. E). Linear probing, which freezes a pretrained encoder and trains only the final linear layer, is a standard diagnostic for representation quality [Alain and Bengio, 2016, Kornblith et al., 2019]. Kumar et al. [2022] show that full fine-tuning can distort strong pretrained features, reducing out-of-distribution accuracy, while last-layer probing preserves feature geometry. Their findings have motivated last-layer strategies in transfer learning. The neural collapse phenomenon [Papayan et al., 2020] further justifies this: near convergence, within-class variability is small and approximately isotropic across classes, a regime in which a LDA head is nearly Bayes-optimal.

A.2 Machine unlearning

The literature is now broad enough that several survey papers cover the field from complementary angles [Nguyen et al., 2022, Xu et al., 2023, Liu et al., 2024, 2025].

Motivation and regulatory context. The renewed interest in machine unlearning is driven by data-protection regulations such as the EU General Data Protection Regulation (GDPR) and its “right to be forgotten” [Voigt and Von dem Bussche, 2017], together with the California Consumer Privacy Act (CCPA) and similar frameworks worldwide. Beyond legal compliance, unlearning has been identified as a tool for removing the influence of poisoned, biased, mislabelled, copyrighted or otherwise undesirable training samples [Cao and Yang, 2015, Bourtole et al., 2021, Goel et al., 2024], and as a primitive for AI safety (e.g., suppressing hazardous knowledge in foundation models) [Li et al., 2024].

Exact unlearning. Machine unlearning was formalised by Cao and Yang [2015], who introduced the term and proposed summation-form algorithms that allow data removal without full retraining. The gold standard remains exact removal, in which the unlearned model is statistically indistinguishable from a model retrained from scratch on the retained set [Cao and Yang, 2015, Bourtole et al., 2021, Sekhari et al., 2021, Ginart et al., 2019]. Ginart et al. [2019] initiated the modern study of approximate deletion in the specific case of k -means clustering, providing the first formal definition of deletion efficiency. Bourtole et al. [2021] achieve exact unlearning in deep models via SISA training, which partitions the data into shards so that only the affected shard needs to be retrained on deletion. Although SISA reduces retraining cost by isolating deletions, it requires storing intermediate checkpoints for every shard, and at scale this storage overhead and the accuracy loss caused by aggressive sharding make it impractical for large deep networks.

Gradient-based approximate unlearning. To avoid retraining, the dominant paradigm relies on gradient-based weight updates. Gradient ascent on the forget-set loss is the simplest baseline and is used as a primitive in several methods [Graves et al., 2021, Chen et al., 2023, Thudi et al., 2022a]. Thudi et al. [2022a] provide an unrolled-SGD analysis of the verification error and show how a training-time penalty can ease subsequent unlearning. Knowledge-distillation variants jointly enforce forgetting and retention: SCRUB [Kurmanji et al., 2023] trains a student that minimises the KL divergence to the original model on the retained data while maximising the loss on the forget set, and Chundawat et al. [2023] use an incompetent teacher whose soft labels guide the student to suppress class knowledge. SalUn [Fan et al., 2024] restricts updates to the weights identified as most responsible for the forgotten data via gradient-based saliency, reaching state-of-the-art results on image classification and generation, and Jia et al. [2023] show that model sparsification prior to unlearning closes the gap between exact and approximate unlearning across multiple algorithms. Tarun et al. [2023] propose UNSIR, an impair repair scheme in which an error-maximising noise first degrades class knowledge and a repair step then restores retain-set accuracy, and Neel et al. [2021] provide gradient-descent-based deletion algorithms that handle adversarial sequences of deletion requests for convex models. Beyond their empirical strengths, these methods share a structural

limitation: the update follows an optimisation trajectory that admits no closed-form characterisation in terms of the data distribution or the resulting predictive shift.

Fisher information and synaptic dampening. Fisher-based methods identify which parameters encode the forgotten data and selectively modify them. Golatkar et al. [2020a] derive a Fisher Information Matrix (FIM) scrubbing procedure that perturbs weights in order to remove cohort information, and Golatkar et al. [2020b] extend this analysis via a Neural Tangent Kernel linearisation that better handles the null-space of the weights and bounds the information accessible from input output observations. Foster et al. [2024] propose Selective Synaptic Dampening (SSD), a retrain-free two-step method that uses the FIMs of the training and forget sets to select and dampen the parameters most important to the forget set. SSD is computationally light and does not require retained data at unlearning time, but its forgetting guarantee is coupled to the quality of the diagonal FIM approximation, which does not expose the structure of the data distribution. Mehta et al. [2022] introduce L-CODEC, a randomised conditional-independence test that selects a Markov blanket of parameters, making Newton-style updates tractable for very large vision and NLP models.

Influence functions and certified removal. Influence functions [Koh and Liang, 2017], borrowed from robust statistics, approximate the leave-one-out parameter shift via a single Newton step on the training loss. Guo et al. [2020] exploit this connection to derive certified data removal for ℓ_2 -regularised linear models: a calibrated noise mechanism masks the residual Newton-step error, yielding an (ϵ, δ) differential-privacy certificate. Sekhari et al. [2021] extend the theoretical analysis by bounding generalisation after unlearning, and Izzo et al. [2021] propose a projection-based deletion algorithm whose cost is linear in the feature dimension and independent of the dataset size for linear and logistic models. Warnecke et al. [2023] extend influence-based unlearning to the deletion of features and labels, providing closed-form updates and certified guarantees for strongly convex losses. Beyond linear models, extending influence-function guarantees to non-convex deep networks remains challenging [Koh and Liang, 2017, Guo et al., 2020]: the Hessian approximation is hard to control, and no closed-form bound on the predictive shift in terms of the data distribution is available. More recent work such as Chien et al. [2024] uses Langevin dynamics to obtain certified unlearning without convexity assumptions, but the resulting bounds still ignore the class-conditional structure of the data.

Weight-space and Bayesian methods. A complementary line of work reasons about model uncertainty in parameter space rather than via optimisation trajectories. Golatkar et al. [2021] extend FIM scrubbing to a mixed-privacy setting by targeting only non-core data, and Peste et al. [2021] scale FIM-based scrubbing to deep networks via rank-one Hessian approximations. The Laplace approximation [Daxberger et al., 2021] provides a principled Gaussian posterior over weights and has been adopted in Bayesian unlearning to characterise the uncertainty introduced by data deletion [Nguyen et al., 2020]. All of these methods operate entirely in weight space: their error guarantees are coupled to Hessian-approximation quality and do not decompose in terms of the quality of the class-conditional data model.

Geometry-based and training-free unlearning. A recent direction exploits the geometric structure of trained representations to obtain closed-form, training-free corrections. Le et al. [2026] build on Neural Collapse theory [Papayan et al., 2020] to show that the orthogonal projection of a simplex Equiangular Tight Frame (ETF) onto the subspace of retained classes remains an ETF in the lower-dimensional space, which yields a provably optimal forgetting operator. Their method, POUR-P, delivers this correction in closed form and without gradient steps. In a similar spirit, Kodge et al. [2024] project model weights onto the subspace orthogonal to the principal directions of the forget concepts, rendering the model blind to that information without retraining. While these methods are fast and elegant, the optimality guarantees of projection-based approaches such as POUR rely on the terminal neural-collapse regime, and neither family provides a data-distribution bound on the residual unlearning error.

Theoretical guarantees and certification criteria. Guo et al. [2020] measure unlearning success as (ϵ, δ) -indistinguishability, mirroring differential privacy [Dwork and Roth, 2014]. Total variation and KL divergence between the ideal retrained model and the empirically unlearned model have since been adopted as natural criteria [Sekhari et al., 2021, Golatkar et al., 2020a, Chien et al., 2024], but nearly all theoretical results are stated in terms of algorithmic properties (Hessian approximation quality, loss Lipschitz constants) rather than in terms of the data distribution. A separate line of work

studies the auditability of unlearning definitions and the difficulty of certifying forgetting in arbitrary training pipelines [Thudi et al., 2022b], underlining the gap between algorithmic and distributional guarantees. To our knowledge, no prior work provides a closed-form decomposition of the expected KL divergence directly in terms of the quality of distributional shift of models fitted to the data (See Prop. 1).

Membership inference as an evaluation tool. Membership inference attacks (MIAs) were introduced by Shokri et al. [2017], formally connected to overfitting by Yeom et al. [2018], and sharpened by the likelihood-ratio attack LiRA of Carlini et al. [2022], which achieves the highest statistical power at low false-positive rates thanks to held-out shadow models. MIAs have become the standard black-box audit for unlearning efficacy [Triantafillou et al., 2024, Hayes et al., 2025]: an effective method should prevent an adversary from distinguishing forgotten examples from held-out non-members. Hayes et al. [2025] show that population-level MIAs systematically overestimate the privacy protection of approximate methods, and they propose U-LiRA, a per-example extension that serves as a stricter worst-case audit. Complementary evaluation protocols have also been developed: Goel et al. [2022] introduce the Interclass Confusion test together with the EU- k and CF- k baselines, and the findings of the first NeurIPS unlearning competition [Triantafillou et al., 2024] provide a broad empirical assessment of the state of the art under a unified evaluation pipeline.

Unlearning beyond classification: generative models and LLMs. Although the present work focuses on classification, machine unlearning has recently been extended to generative models, where it is used to suppress copyrighted, sensitive or harmful content. For diffusion models, Gandikota et al. [2023] propose Erased Stable Diffusion (ESD), a fine-tuning method that removes a target concept from the model weights using only a textual description and negative guidance. SalUn [Fan et al., 2024] also covers concept erasure in diffusion models. For large language models, Eldan and Russinovich [2023] introduce the “Who’s Harry Potter?” protocol for unlearning copyrighted content, Maini et al. [2024] propose the TOFU benchmark of fictitious authors for controlled evaluation, and Li et al. [2024] introduce the WMDP benchmark for hazardous-knowledge unlearning together with the RMU representation misdirection method. These efforts highlight that distributional unlearning guarantees are even more elusive in the generative regime, since the relevant “data distribution” is itself implicit in the model.

Federated and distributed unlearning. A parallel literature studies unlearning in the federated setting, where clients hold local data and only model updates are shared with a central server. FedEraser [Liu et al., 2021] reconstructs the global model from cached client updates while excluding the contribution of a target client, and a growing number of follow-up methods address client-level and sample-level deletion under heterogeneous data and asynchronous participation; Romandini et al. [2024] provide a comprehensive survey of methods, design guidelines and evaluation metrics. While these contributions are largely orthogonal to ours, the federated setting introduces communication and asynchronicity constraints absent in the centralised case, they share with the centralised literature the property that error guarantees are stated in terms of algorithmic quantities rather than data-distribution quality.

B Glossary

This appendix gathers in a single place all the notations introduced throughout the paper. The challenge with our framework is that several layers of probability distributions co-exist: the unknown true distribution, the neural network’s predictive distribution, and the parametric mathematical proxies. Table 4 lists the symbols with a short definition and an indication of where it is introduced in the paper.

Notation	Definition
$\mathcal{X}$	Input space (e.g. images, embeddings).
$\mathcal{Y} = \llbracket C \rrbracket$	Label space; C is the number of classes.
$\mathcal{D}$	Original training set, an i.i.d. sample from $\mathbb{P}$.
$\mathcal{D}_r$	<i>Retain set</i> : data to be preserved, sampled from $\mathbb{P}_r$.
$\mathcal{D}_f$	<i>Forget set</i> : data whose influence must be erased; $\mathcal{D} = \mathcal{D}_r \cup \mathcal{D}_f$.
$\mathcal{D}(y), \mathcal{D}_r(y), \mathcal{D}_f(y)$	Class- y subsets of the corresponding sets.
$ \mathcal{D}_f , \mathcal{D}_r , \mathcal{D} $	Sizes of sets $\mathcal{D}_f, \mathcal{D}_r, \mathcal{D}$.
π_r, π_f	Mixture weights of the retain/forget components in $\mathbb{P}$ (A2); $\pi_r + \pi_f = 1$.
$\pi_f(y)$	Intra-class proportion $ \mathcal{D}_f(y) / \mathcal{D}(y) $ used in the conditional mixture (13).
$s \in \{r, f\}$	State variable that doubles the labels in the $2C$ construction (Sect. 4).
$\mathbb{P}$	True (unknown) joint distribution of (X, Y) . $\mathcal{D}$ is i.i.d. from $\mathbb{P}$ (Assumption A1).
$\mathbb{P}_X, \mathbb{P}_Y$	Marginal true distributions.
$\mathbb{P}_{Y X}, \mathbb{P}_{X Y}$	Conditional true distributions.
$\mathbb{P}_r$	True distribution of the retain set; $\mathcal{D}_r \sim \mathbb{P}_r^{ \mathcal{D}_r }$.
$\mathbb{P}_f$	True distribution of the forget component in the mixture $\mathbb{P} = \pi_r \mathbb{P}_r + \pi_f \mathbb{P}_f$.
$\mathbf{f}_\theta(x) \in \mathbb{R}^C$	Logit vector output of the initial neural network at input x .
$\hat{\mathbf{f}}_\eta(x), \tilde{\mathbf{f}}(x)$	Logit vector of our unlearned NN; $\hat{\mathbf{f}}_\eta = \mathbf{f}_\theta + \eta \Delta \mathbf{M}$ for $\eta \in [0, \eta_{\max}]$. The final model uses $\eta = \eta_{\max}$.
$\mathbf{p}_\theta(x)$	Predicted probabilities (probits) of the initial NN: $\mathbf{p}_\theta(x) \doteq \text{softmax}(\mathbf{f}_\theta(x))$.
$\mathbf{p}_{\theta_r}(x)$	Predicted probabilities (probits) of the retrained NN.
$\tilde{\mathbf{p}}(x)$	Probits of the unlearned model: $\tilde{\mathbf{p}}(x) = \text{softmax}(\tilde{\mathbf{f}}(x))$.
$\mathbf{p}_{\tilde{\theta}}(x)$	Probits of the unlearned NN after distillation.
$\mathbf{P}(x y)$	Initial proxy: a parametric model of $\mathbb{P}_{X Y}$ fitted on $\mathcal{D}$.
$\mathbf{P}_r(x y)$	Retrained proxy: a parametric model of $\mathbb{P}_{r,X Y}$ fitted on $\mathcal{D}_r$ or recovered from a $2C$ construction.
$\mathbf{P}_f(x y)$	Per-class proxy of the forget distribution used in the conditional mixture (13).
$\mathbf{P}(y x), \mathbf{P}_r(y x)$	Posteriors induced by the proxies via Bayes rule (1).
$\mathbf{M}(x), \mathbf{M}_r(x) \in \mathbb{R}^C$	Proxy logits: $\mathbf{M}(x)_y \doteq \log \mathbf{P}_{Y X}(y x)$ (idem $\mathbf{M}_r$).
$\Delta \mathbf{M}(x)$	$\Delta \mathbf{M}(x) \doteq \mathbf{M}_r(x) - \mathbf{M}(x)$ is the unlearning signal, also referred as proxy shift.
softmax	Standard softmax operator $\mathbb{R}^C \rightarrow \mathcal{S}_C$.
$\text{LSE}(z)$	Log-Sum-Exponential: $\text{LSE}(z) \doteq \log \sum_{i=1}^C e^{z_i}$.
$\gamma(x)$	Normalization constant of the initial NN: $\gamma(x) \doteq \text{LSE}(\mathbf{f}_\theta(x))$, so that $\mathbf{f}_\theta(x) = \log(\mathbf{p}_\theta(x)) + \gamma(x)\mathbf{1}$.
$\tilde{\gamma}(x; \eta)$	Normalization of the unlearned model: $\tilde{\gamma}(x; \eta) \doteq \text{LSE}(\tilde{\mathbf{f}}_\eta(x))$.
$\Delta \gamma(x; \eta)$	Shift of the normalizations between the unlearned and the initial NN. It is convex in η , with $\Delta \gamma(x; 0) = 0$ and its expectation w.r.t. $\mathbb{P}$ is negative under A4.
$\eta \in [0, 1]$	Scaling parameter of the unlearning signal.
$\eta_{\max}$	Largest $\eta > 0$ such that $\mathbb{E} \Delta \gamma(X; \eta) < 0$; defines our theoretically safe regime. Found by 1-D bisection.
$\mathbf{1}$	All-one vector in $\mathbb{R}^C$.
$\text{KL}_{Y x}(p q)$	$\sum_{y=1}^C p_{Y x}(y) \log \frac{p_{Y x}(y)}{q_{Y x}(y)}$. The KL divergence between conditional distributions.
$\mathbb{E} \text{KL}_{Y X}(p q)$	Its expectation over $X \sim \mathbb{P}_X$. Main figure of merit of the paper (Sect. 3.4).
$\mathbb{E} \text{KL}_{X Y}(p q)$	Symmetric expression conditioning on Y , used in the bound of Prop. 2 on raw class-conditional modeling errors.
$H(p)$	Shannon entropy of p .

Table 4: List of the symbols with their short definition.

C Justifications

C.1 The unlearned model family

We look for a model $\tilde{p}$, which is an update similar to the initial NN p_θ , closer to the ideal retain distribution $\mathbb{P}_r$ than the initial distribution $\mathbb{P}$. This objective is framed in terms of KL divergences in the following optimization problem:

$$\min_{\tilde{p}} \mathbb{EKL}_{Y|X}(\tilde{p}||p_\theta) \quad \text{s.t.} \quad \mathbb{EKL}_{Y|X}(\tilde{p}||\mathbb{P}_r) \leq \mathbb{EKL}_{Y|X}(\tilde{p}||\mathbb{P}), \quad \mathbb{E} \sum_y \tilde{p}(y|X) = 1. \quad (19)$$

An interpretation is the following. Among all distributions $\tilde{p}$ that fall on the $\mathbb{P}_r$ side of the decision boundary (discriminating whether samples from $\tilde{p}$ are more likely to be sampled from $\mathbb{P}$ or $\mathbb{P}_r$), we seek the one closest to p_θ in relative entropy *i.e.* the most p_θ -like distribution that the Neyman-Pearson likelihood test would still attribute in expectation to $\mathbb{P}_r$ instead of $\mathbb{P}$.

By definition of the KL one always has the identity

$$\mathbb{EKL}_{Y|X}(\tilde{p}||\mathbb{P}) - \mathbb{EKL}_{Y|X}(\tilde{p}||\mathbb{P}_r) = \mathbb{E} \sum_y \tilde{p}(y|X) \log \frac{\mathbb{P}_r(y|X)}{\mathbb{P}(y|X)}.$$

Following [Cover and Thomas, 2006, Chap. 12], we write the Lagrangian with multipliers $\eta \geq 0, \nu$ associated to (19) as

$$J(\tilde{p}; \eta, \nu) \doteq \mathbb{E} \sum_y \tilde{p}(y|X) \log \frac{\tilde{p}(y|X)}{p_\theta(y|X)} - \eta \mathbb{E} \sum_y \tilde{p}(y|X) \log \frac{\mathbb{P}_r(y|X)}{\mathbb{P}(y|X)} + \nu \mathbb{E} \sum_y \tilde{p}(y|X).$$

Cancelling the derivative with respect to $\tilde{p}(y|X)$, we obtain: $\forall y \in \mathcal{Y}$

$$\mathbb{E} \log \frac{\tilde{p}(y|X)}{p_\theta(y|X)} + 1 - \eta \mathbb{E} \log \frac{\mathbb{P}_r(y|X)}{\mathbb{P}(y|X)} + \nu = 0. \quad (20)$$

Rearranging the terms, the final expression can be expressed as:

$$\mathbb{E} \left[\underbrace{\log(\tilde{p}_\eta(\cdot|X))}_{\tilde{f}(X) - \tilde{\gamma}(X; \eta) \mathbb{1}} \right] = \mathbb{E} \left[\underbrace{\log p_\theta(\cdot|X)}_{f_\theta(X) - \gamma(X) \mathbb{1}} + \underbrace{\log \frac{\mathbb{P}_r(\cdot|X)}{\mathbb{P}(\cdot|X)}}_{\text{exact unlearning signal } \Delta \mathbb{M}(X)} - \underbrace{\Delta \gamma(X; \eta)}_{\text{normalizing term}} \right]. \quad (21)$$

Remark In their work Cover and Thomas [2006], $\eta = \eta_0$ is chosen precisely on the attacker decision boundary :

$$\mathbb{EKL}_{Y|X}(\tilde{p}_{\eta_0}||\mathbb{P}) - \mathbb{EKL}_{Y|X}(\tilde{p}_{\eta_0}||\mathbb{P}_r) = 0.$$

so that an attacker sampling from $\tilde{p}$ is not, in expectation, in capacity to decide whether the drawn samples are coming from $\mathbb{P}_r$ or $\mathbb{P}$. By the symmetry $\eta \longleftrightarrow 1 - \eta$, the same $\tilde{p}_\eta$ minimizes $\mathbb{EKL}_{Y|X}(\tilde{p}||\mathbb{P}_r)$ constrained to the reversed inequality.

Estimation One solution is to impose that (21) holds for any x . This translates in the logit space to the solution:

$$\tilde{f}(x) = f_\theta(x) + \eta \Delta \mathbb{M}(x),$$

a.k.a. precisely our modeling of the unlearning family in Section 3. The true exact unlearning signal $\Delta \mathbb{M}(x) = \log \mathbb{P}_r(\cdot|X) / \mathbb{P}(\cdot|X)$ being inaccessible in practice, our key idea consists in creating proxies (P, P_r) that approximate this quantity.

$$\Delta \mathbb{M} \approx \Delta \mathbb{M}$$

Rephrasing, given proxies (P, P_r) we obtain our unlearned model family as

$$\tilde{f}(x) = f_\theta(x) + \eta \Delta \mathbb{M}(x)$$

More precisely, it is the unbiased solution of the following optimization problem:

$$\min_{\tilde{p}} \mathbb{EKL}_{Y|X}(\tilde{p}||p_\theta) \quad \text{s.t.} \quad \mathbb{EKL}_{Y|X}(\tilde{p}||P_r) \leq \mathbb{EKL}_{Y|X}(\tilde{p}||P), \quad \mathbb{E} \sum_y \tilde{p}(y|X) = 1.$$

To conclude, we consider the following family of unlearned models:

$$\tilde{f}_\eta(x) \doteq f_\theta(x) + \eta \Delta M(x), \quad \Delta M(x) \doteq M_r(x) - M(x). \quad (22)$$

As introduced in Sect. 3.1, logits and probits are related by:

$$\tilde{p}_\eta(x) \doteq \text{softmax}(\tilde{f}_\eta(x)), \quad \tilde{f}_\eta(x) = \log(\tilde{p}_\eta(x)) + \tilde{\gamma}(x; \eta). \quad (23)$$

C.2 The safety range of η

We first relate the normalization $\tilde{\gamma}(x; \eta)$ to $\gamma(x)$, the normalization of the initial NN.

$\text{LSE} : \mathbb{R}^C \rightarrow \mathbb{R}$ denotes the Log-Sum-Exponential function defined as $\text{LSE}(z) \doteq \log \left(\sum_{i=1}^C e^{z_i} \right)$.

It verifies, $\forall z \in \mathbb{R}^C, \forall \kappa \in \mathbb{R}$:

$$\log(\text{softmax}(z)) = z - \text{LSE}(z) \mathbb{1} \quad (24)$$

$$\text{LSE}(z + \kappa \mathbb{1}) = \text{LSE}(z) + \kappa \quad (25)$$

$$\text{LSE}(\log(z)) = \log \left(\sum_{i=1}^C z_i \right) \quad (26)$$

where $\log(z)$ is an abuse of notation defining the vector $(\log z_1, \dots, \log z_C)$ for $z \in \mathbb{R}_{>0}^C$, and $\mathbb{1} \in \mathbb{R}^C$ is the all-one vector.

Indeed, applying LSE to the logits of the unlearned model $\tilde{f}_\eta(x) = \log p_{\tilde{\theta}_\eta}(x) + \tilde{\gamma}(x; \eta) \mathbb{1}$, isolates the normalization term due to (25) and (26): $\text{LSE}(\tilde{f}_\eta(x)) = \tilde{\gamma}(x; \eta)$. Since we forge the unlearned model by adding the proxy shift to the initial NN (2), we have:

$$\begin{aligned} \tilde{\gamma}(x; \eta) &\stackrel{(22)}{=} \text{LSE} \left(f_\theta(x) + \eta \log \left(\frac{P_r(\cdot|x)}{P(\cdot|x)} \right) \right) \\ &\stackrel{(23)}{=} \text{LSE} \left(\log p_\theta(x) + \gamma(x) \mathbb{1} + \eta \log \left(\frac{P_r(\cdot|x)}{P(\cdot|x)} \right) \right) \\ &\stackrel{(25)}{=} \gamma(x) + \text{LSE} \left(\log \left(\frac{p_\theta(x) P_r^\eta(\cdot|x)}{P^\eta(\cdot|x)} \right) \right) \\ &\stackrel{A3}{=} \gamma(x) + \text{LSE} \left(\log \left(\frac{\mathbb{P}(x) P_r^\eta(\cdot|x)}{P^\eta(\cdot|x)} \right) \right), \end{aligned}$$

with component-wise multiplication and division between vectors. This links the normalization offset $\tilde{\gamma}$ of the unlearned model to γ , the one of the initial model. Thanks to (26), we have:

$$\begin{aligned} \Delta\gamma(x; \eta) &\doteq \tilde{\gamma}(x; \eta) - \gamma(x) = \log \left(\sum_{y=1}^C \mathbb{P}(y|x) \left(\frac{P_r(y|x)}{P(y|x)} \right)^\eta \right) \\ &= \log \left(\mathbb{E}_{Y|x \sim \mathbb{P}} \left[\left(\frac{P_r(Y|x)}{P(Y|x)} \right)^\eta \right] \right). \end{aligned}$$

Note that $\Delta\gamma(x; 0) = 0, \forall x$, which is expected as the unlearning operator is void at $\eta = 0$. Its derivative w.r.t. η equals

$$\frac{\partial \Delta\gamma(x; \eta)}{\partial \eta} = \frac{\mathbb{E}_{Y|x \sim \mathbb{P}} [A^\eta(Y, x) \log(A(Y, x))]}{\mathbb{E}_{Y|x \sim \mathbb{P}} [A^\eta(Y, x)]},$$

with $A(Y, x) \doteq P_r(Y|x)/P(Y|x)$, and because the derivative of $a^x = \exp(x \log(a))$ equals $a^x \log(a)$, $\forall a > 0$. Note that, at point $\eta = 0$,

$$\begin{aligned} \left. \frac{\partial \Delta\gamma(x; \eta)}{\partial \eta} \right|_{\eta=0} &= \mathbb{E}_{Y|x \sim \mathbb{P}} \left[\log \left(\frac{P_r(Y|x)}{P(Y|x)} \right) \right] \\ &= \text{KL}_{Y|x}(\mathbb{P}||\mathbb{P}) - \text{KL}_{Y|x}(\mathbb{P}||P_r). \end{aligned}$$

On expectation over $\mathbb{P}_X$, we obtain:

$$\mathbb{E} \left[\left. \frac{\partial \Delta \gamma(X; \eta)}{\partial \eta} \right|_{\eta=0} \right] = \mathbb{E} \text{KL}_{Y|X}(\mathbb{P} \| \mathbb{P}) - \mathbb{E} \text{KL}_{Y|X}(\mathbb{P} \| \mathbb{P}_r) \stackrel{\text{A4}}{\leq} 0,$$

if $\mathbb{P}$ is a better proxy of $\mathbb{P}$ than $\mathbb{P}_r$ (see Sect. 3.3, assumption A4). This assumption is verified experimentally in Sect. 5.2.

In the same way, the second derivative verifies, $\forall x \in \mathcal{X}$:

$$\frac{\partial^2 \Delta \gamma(x; \eta)}{\partial^2 \eta} = \frac{\mathbb{E}_{Y|x \sim \mathbb{P}} [A^\eta(Y, x) \log^2(A(Y, x))]}{E_{Y|x \sim \mathbb{P}} [A^\eta(Y, x)]} - \frac{\mathbb{E}_{Y|x \sim \mathbb{P}} [A^\eta(Y, x) \log(A(Y, x))]^2}{(E_{Y|x \sim \mathbb{P}} [A^\eta(Y, x)])^2} \geq 0$$

thanks to the Cauchy-Schwarz inequality.

To recap, $\Delta \gamma(\eta) \doteq \mathbb{E} [\Delta \gamma(X; \eta)]$ enjoys the following properties:

- $\Delta \gamma(0) = 0$
- $\Delta \gamma'(0) \leq 0$
- $\Delta \gamma(\eta)$ is convex over $[0, +\infty)$.

Consequently, there exist $\eta_{\max} > 0$ such that $\Delta \gamma(\eta) \leq 0$ for all $0 \leq \eta \leq \eta_{\max}$.

In practice, $\eta_{\max}$ is found by a line search as the zero of the function:

$$h(\eta) = \frac{1}{N} \sum_{x \in \mathcal{D}} \text{LSE}(f_\theta(x) + \eta \Delta M(x)) - \text{LSE}(f_\theta(x)) \simeq \Delta \gamma(\eta). \quad (27)$$

Remark The careful reader will spot that the derivative of $\Delta \gamma$ in expectation cancels out exactly at η_0 : the regime of indistinguishability between $\mathbb{P}$ and $\mathbb{P}_r$. Then $\eta_0 \leq \eta_{\max}$ as the negative derivative hypothesis indicates. Taking $\eta = \eta_{\max}$ ensures we have been through the equality boundary and we are in the $\mathbb{E} \text{KL}_{Y|X}(\tilde{\mathbb{P}}_\eta \| \mathbb{P}) > \mathbb{E} \text{KL}_{Y|X}(\tilde{\mathbb{P}}_\eta \| \mathbb{P}_r)$ regime.

D Main results

D.1 Study of the logits error and proof of Prop. 3

We now define the error of prediction between the true distribution $\mathbb{P}_{r,Y|x}$ and the unlearned model $\tilde{\mathbb{P}}_\eta$, in log scale:

$$\mathcal{E}(y, x; \eta) \doteq \log(\mathbb{P}_r(y|x)) - \log(\tilde{\mathbb{P}}_\eta(y|x)).$$

This quantity is worth studying as it appears in the computation of $\mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| \tilde{\mathbb{P}}_\eta)$, the quantity of importance in this paper. We have:

$$\text{KL}_{Y|X}(\mathbb{P}_r \| \tilde{\mathbb{P}}_\eta) = \sum_{y=1}^C \mathbb{P}_r(y|x) \log \left(\frac{\mathbb{P}_r(y|x)}{\tilde{\mathbb{P}}_\eta(y|x)} \right) = \langle \mathbb{P}_r(\cdot|x), \mathcal{E}(\cdot, x, \eta) \rangle. \quad (28)$$

On the other hand, we have

$$\begin{aligned} \mathcal{E}(x, y; \eta) &= \log(\mathbb{P}_r(y|x)) - \tilde{f}_\eta(x) + \tilde{\gamma}(x; \eta) \\ &\stackrel{(2)}{=} \log(\mathbb{P}_r(y|x)) - f(x) - \eta \Delta M(x) + \tilde{\gamma}(x; \eta) \\ &= \log(\mathbb{P}_r(y|x)) - \log p_\theta(y|x) - \gamma(x) - \eta \Delta M(x) + \tilde{\gamma}(x; \eta) \\ &= \log \left(\frac{\mathbb{P}_r(y|x)}{p_\theta(y|x)} \right) - \eta \log \left(\frac{\mathbb{P}_r(y|x)}{\mathbb{P}(y|x)} \right) + \Delta \gamma(x; \eta) \end{aligned} \quad (29)$$

Plugging (29) into (28) and taking the expectation leads to:

$$\begin{aligned} \mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| \tilde{\mathbb{P}}_\eta) &= \mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| p_\theta) \\ &\quad + \underbrace{\mathbb{E} [\Delta \gamma(X, \eta)]}_{< 0, \forall \eta \in [0, \eta_{\max}]} + \eta \left(\underbrace{\mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}_r) - \mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| \mathbb{P})}_{< 0 \text{ if } (\mathbb{P}_r, \mathbb{P}) \text{ is } \mathbb{P}_r\text{-admissible}} \right). \end{aligned}$$

The first term is the KL divergence without unlearning, *i.e.* between the true retain distribution and the initial NN probits. App. C.2 shows that the second term is negative under A4 (5) for any $0 \leq \eta \leq \eta_{\max}$. The third term is negative with one extra assumption: if (P_r, P) is $\mathbb{P}_r$ -admissible, *i.e.* P_r is a better proxy for $\mathbb{P}_r$ than P is.

This proves Prop. 3.

D.2 Proof of Prop. 1

We rewrite (29) in order to obtain another expression of the error $\mathcal{E}$:

$$\mathcal{E}(x, y; \eta) \stackrel{A3}{=} (1 - \eta) \log \left(\frac{\mathbb{P}_r(y|x)}{\mathbb{P}(y|x)} \right) + \eta \left(\log \left(\frac{\mathbb{P}_r(y|x)}{\mathbb{P}_r(y|x)} \right) - \log \left(\frac{\mathbb{P}(y|x)}{\mathbb{P}(y|x)} \right) \right) + \Delta\gamma(x; \eta)$$

We now take the expectation w.r.t. $\mathbb{P}_{rY|x}$ and then the expectation w.r.t. $\mathbb{P}_X$. As earlier, we assume A4 (5) holds and $\eta \in [0, \eta_{\max}]$ so that $\Delta\gamma(\eta)$ is non-positive. This leads to:

$$\mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \tilde{P}_\eta) \leq (1 - \eta) \mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}) + \eta \left(\mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}_r) - \underbrace{\mathbb{E} \sum_{y=1}^C \mathbb{P}_r(y|X) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)}}_{\text{term } A} \right)$$

Note that we get an equality when $\eta = \eta_{\max}$ because $\Delta\gamma(\eta_{\max}) = 0$ by definition of $\eta_{\max}$. The following lemma helps us work out term A .

Lemma 1 (Formal Calculus). *According to assumption A2 (4) about the true distributions, $\mathbb{P} = (1 - \pi_f)\mathbb{P}_r + \pi_f\mathbb{P}_f$ over the joint data (X, Y) . Then we have the following formal identity with respect to X, Y or (X, Y) :*

$$\mathbb{E} = \mathbb{E}_r + \pi_f (\mathbb{E}_r - \mathbb{E}_f). \quad (30)$$

Applying this lemma to the term A :

$$A \stackrel{(30)}{=} \mathbb{E}_r[Z] + \pi_f \left(\mathbb{E}_r[Z] - \mathbb{E}_f \sum_{y=1}^C \mathbb{P}_r(y|X) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)} \right),$$

with $Z = \log(\mathbb{P}(Y|X)/\mathbb{P}(Y|X))$.

$$\begin{aligned} A &\stackrel{(30)}{=} \mathbb{E}[Z] - \pi_f (\mathbb{E}_r[Z] - \mathbb{E}_f[Z]) + \pi_f \left(\mathbb{E}_r[Z] - \mathbb{E}_f \sum_{y=1}^C \mathbb{P}_r(y|X) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)} \right) \\ &= \mathbb{E}[Z] + \pi_f \left(\mathbb{E}_f[Z] - \mathbb{E}_f \sum_{y=1}^C \mathbb{P}_r(y|X) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)} \right) \\ &= \mathbb{E} \left[\log \frac{\mathbb{P}(Y|X)}{\mathbb{P}(Y|X)} \right] + \pi_f \mathbb{E}_f \left[\sum_{y=1}^C \left((\mathbb{P}_f(y|X) - \mathbb{P}_r(y|X)) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)} \right) \right]. \end{aligned}$$

The second term is not really tractable and will be referred to as $\mathcal{O}(\pi_f)$.

$$\mathcal{O}(\pi_f) \doteq \pi_f \mathbb{E}_f \left[\sum_{y=1}^C \left((\mathbb{P}_f(y|X) - \mathbb{P}_r(y|X)) \log \frac{\mathbb{P}(y|X)}{\mathbb{P}(y|X)} \right) \right].$$

The constant is close to zero either when $\mathbb{P}_f(y|X)$ and $\mathbb{P}_r(y|X)$ take similar values (no unlearning signal), or when the initial proxy is close to the initial true distribution (*c.f.* good proxy assumption).

In the end, we obtain

$$\mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \tilde{P}_\eta) \leq (1 - \eta) \mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}) + \eta (\mathbb{EKL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}_r) - \mathbb{EKL}_{Y|X}(\mathbb{P} \| \mathbb{P})) + \mathcal{O}(\pi_f).$$

This proves Prop. 1.

D.3 Proof of Prop. 2

This section converts the previous bound with probabilities $Y|X$ into terms modeling $X|Y$ thanks to the following lemma.

Lemma 2. *For any distributions $\mathbb{P}$ and $\mathbb{Q}$ over (X, Y) with the same marginal $\mathbb{P}_Y = \mathbb{Q}_Y$:*

$$\mathbb{E}_{X \sim \mathbb{P}} [\text{KL}_{Y|X} (\mathbb{P} \parallel \mathbb{Q})] = \mathbb{E}_{Y \sim \mathbb{P}} [\text{KL}_{X|Y} (\mathbb{P} \parallel \mathbb{Q})] - \text{KL}_X (\mathbb{P} \parallel \mathbb{Q}) \quad (31)$$

Proof.

$$\begin{aligned} \mathbb{E}_{X \sim \mathbb{P}} [\text{KL}_{Y|X} (\mathbb{P} \parallel \mathbb{Q})] &= \mathbb{E}_{X \sim \mathbb{P}} \left[\mathbb{E}_{Y \sim \mathbb{P}_{Y|X}} \left[\log \frac{\mathbb{P}(Y|X)}{\mathbb{Q}(Y|X)} \right] \right] \\ &= \mathbb{E}_{(X,Y) \sim \mathbb{P}} \left[\log \frac{\mathbb{P}(Y|X)}{\mathbb{Q}(Y|X)} \right] \\ &\stackrel{\text{Bayes}}{=} \mathbb{E}_{(X,Y) \sim \mathbb{P}} \left[\log \frac{\mathbb{P}(X|Y) \mathbb{Q}(X)}{\mathbb{Q}(X|Y) \mathbb{P}(X)} \right] \\ &= \mathbb{E}_{Y \sim \mathbb{P}} [\text{KL}_{X|Y} (\mathbb{P} \parallel \mathbb{Q})] - \text{KL}_X (\mathbb{P} \parallel \mathbb{Q}), \end{aligned}$$

because $\mathbb{E}_{(X,Y) \sim \mathbb{P}} \left[\log \frac{\mathbb{P}(X)}{\mathbb{Q}(X)} \right] = \mathbb{E}_{X \sim \mathbb{P}} \left[\log \frac{\mathbb{P}(X)}{\mathbb{Q}(X)} \right] = \text{KL}_X (\mathbb{P} \parallel \mathbb{Q})$ considers only the marginal distributions of X . $\square$

Applying this lemma to the second term of the previous bound to leads to:

$$\begin{aligned} \mathbb{E} \text{KL}_{Y|X} (\mathbb{P}_r \parallel \mathbb{P}_r) - \mathbb{E} \text{KL}_{Y|X} (\mathbb{P} \parallel \mathbb{P}) &\stackrel{(31)}{=} \mathbb{E} \text{KL}_{X|Y} (\mathbb{P}_r \parallel \mathbb{P}_r) - \mathbb{E} \text{KL}_X (\mathbb{P}_r \parallel \mathbb{P}_r) \\ &\quad - \mathbb{E} \text{KL}_{X|Y} (\mathbb{P} \parallel \mathbb{P}) + \mathbb{E} \text{KL}_X (\mathbb{P} \parallel \mathbb{P}) \\ &\leq \mathbb{E} \text{KL}_{X|Y} (\mathbb{P}_r \parallel \mathbb{P}_r) + \mathbb{E} \text{KL}_X (\mathbb{P} \parallel \mathbb{P}), \end{aligned}$$

because KL divergences are non-negative. In the end,

$$\begin{aligned} \mathbb{E} \text{KL}_{Y|X} (\mathbb{P}_r \parallel \tilde{\mathbb{P}}_\eta) &\leq (1 - \eta) \mathbb{E} \text{KL}_{Y|X} (\mathbb{P}_r \parallel \mathbb{P}) \\ &\quad + \eta (\mathbb{E} \text{KL}_{X|Y} (\mathbb{P}_r \parallel \mathbb{P}_r) + \text{KL}_X (\mathbb{P} \parallel \mathbb{P})) + \mathcal{O}(\pi_f) \end{aligned}$$

Conditioning increases divergence, *i.e.* $\text{KL}_X (\mathbb{P} \parallel \mathbb{P}) \leq \mathbb{E} \text{KL}_{X|Y} (\mathbb{P} \parallel \mathbb{P})$ since $\mathbb{P}_Y = \mathbb{P}_Y$ (see (1)), so that:

$$\mathbb{E} \text{KL}_{Y|X} (\mathbb{P}_r \parallel \tilde{\mathbb{P}}_\eta) \leq \eta (\mathbb{E} \text{KL}_{X|Y} (\mathbb{P}_r \parallel \mathbb{P}_r) + \mathbb{E} \text{KL}_{X|Y} (\mathbb{P} \parallel \mathbb{P})) + \mathcal{O}(\pi_f) + \mathcal{O}(1 - \eta)$$

This proves Prop. 2.

D.4 Assumption A3

This section quantifies the additional error term obtained when assumption (A3) is violated. In terms of logits, the relaxation of the assumptions gives

$$\text{softmax}(\log \tilde{\mathbb{P}}_\eta) = \text{softmax}(\log \mathbb{P} + \eta \Delta \mathbb{M} + \epsilon) \quad (32)$$

with $\epsilon = \log p_{\theta\theta} - \log \mathbb{P}$. One can rewrite

$$\mathbb{E} \text{KL}_{Y|X} (\mathbb{P}_r \parallel \tilde{\mathbb{P}}_\eta) = \mathbb{E} \text{KL}_{Y|X} \left(\mathbb{P}_r \parallel \underbrace{\text{softmax}(\log \mathbb{P} + \eta \Delta \mathbb{M} + \epsilon)}_{(A3)} \right) = \mathbb{E} \text{KL}_{Y|X} \left(\mathbb{P}_r \parallel \tilde{\mathbb{P}}_\eta^{(A3)} \right) + C(\epsilon) \quad (33)$$

where $\tilde{\mathbb{P}}_\eta^{(A3)}$ is what would be $\tilde{\mathbb{P}}_\eta$ assuming (A3), *i.e.* when $\epsilon = 0$. Dymetman [2026] explicitly computes of the cost $C(\epsilon)$ in the context of exponential families.

E About discriminant analysis proxies

Homoscedastic Gaussian data. Linear Discriminant Analysis (LDA) assumes a generative model where each class follows a Gaussian distribution with a shared covariance:

$$x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma). \quad (34)$$

The posterior probability has the following expression [McLachlan, 1992]:

$$\mathbb{P}(y = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_{j=1}^C \exp(w_j^\top x + b_j)},$$

with row weights $w_k = \Sigma^{-1}\mu_k$ and biases $b_k = \ln \pi_k - \frac{1}{2}w_k\mu_k$, where π_k is the class prior. The empirical moment $(\hat{\pi}_k, \hat{\mu}_k, \hat{\Sigma})$ estimated from $\mathcal{D}$ yield the Maximum Likelihood Estimator which exactly has the form of a linear layer ($W_{\text{LDA}} = (w_1, \dots, w_C)^\top$, $b_{\text{LDA}} = (b_1, \dots, b_C)^\top$) followed by the softmax operator. Indeed, as $n \rightarrow \infty$, this MLE coincides with the global minimizer of the cross-entropy loss over any linear model, making LDA the Bayes-optimal classifier in this setting [McLachlan, 1992, Pappayan et al., 2020]. This explains the success of last-layer linear probing on strongly pre-trained features [Kumar et al., 2022, Alain and Bengio, 2016]: when frozen features are approximately class-conditional Gaussian distributed, fitting a linear head by cross-entropy is equivalent to computing the MLE in closed form.

Heteroscedastic Gaussian data. When the homoscedasticity assumption is relaxed, each class k has its own covariance Σ_k :

$$x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma_k). \quad (35)$$

This is Quadratic Discriminant Analysis (QDA). Define the quadratic discriminant as

$$Q_k(x) = -\frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1}(x - \mu_k) - \frac{1}{2} \ln |\Sigma_k| + \ln \pi_k, \quad (36)$$

so that the posterior probability $\mathbb{P}(y = k \mid x) = \text{softmax}(\exp(Q(x)))_k$ [McLachlan, 1992]. Under the heteroscedastic Gaussian assumption and in the limit $n \rightarrow \infty$, minimizing the cross-entropy loss drives the neural network’s predictive distribution to this softmax [McLachlan, 1992].

Even when the Gaussian assumption is only approximate, LDA and QDA remain powerful and computationally efficient proxies as confirmed by the strong linear-probing results of recent foundation models [Oquab et al., 2024, Kumar et al., 2022]. Figure 3 illustrates this equivalence on a toy example.

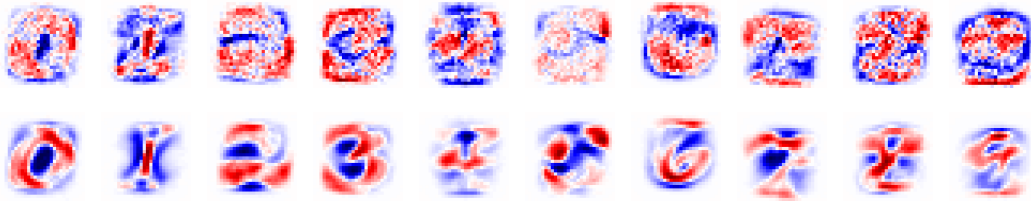


Figure 3: **Cross-entropy training matches LDA.** Top: last-layer weight matrix (rows reshaped as images) of a linear network trained with the Adam optimizer and cross-entropy loss. Bottom: closed-form LDA weight formula. Dataset: MNIST [LeCun et al., 1998]. The near-perfect visual match empirically confirms the cross-entropy/LDA equivalence.

Cost of Homoscedastic Assumption This subsection exhibits the cost to model the retain proxy by LDA rather than QDA.

Theorem 1 (Pythagorean Theorem of Information Csiszár [1975]). *Let $\mathcal{P}$ be a closed convex set of probability distributions, $p_0 \notin \mathcal{P}$, and $p^* = \arg \min_{p \in \mathcal{P}} \text{KL}(p \| p_0)$ the I-projection of p_0 onto $\mathcal{P}$. Then for any $p \in \mathcal{P}$:*

$$\text{KL}(p \| p_0) \geq \text{KL}(p \| p^*) + \text{KL}(p^* \| p_0), \quad (37)$$

with equality when $\mathcal{P}$ is a linear family.

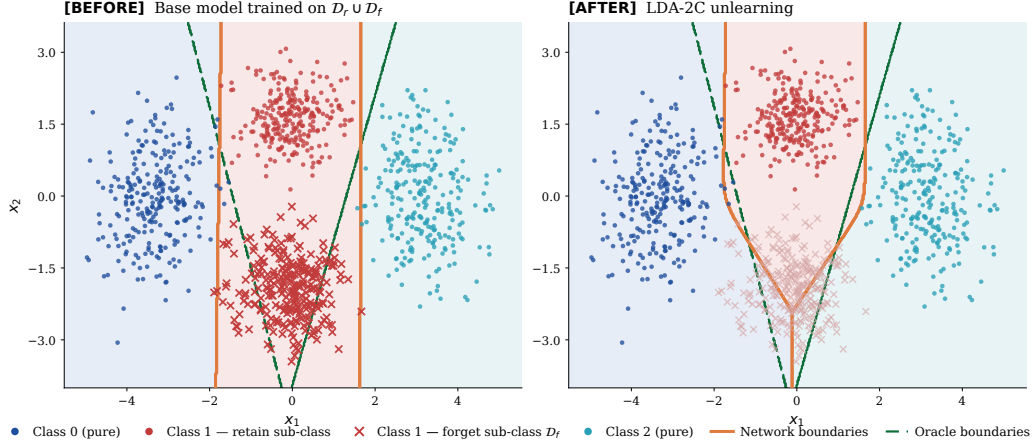


Figure 4: Illustration of the sub-class scenario over synthetic heteroscedastic Gaussian data. Class 1 is composed of two sub-classes, green and red points. The latter have to be unlearned.

We apply this theorem class by class conditioning on $Y = k$ for $k \in \{1, \dots, C\}$. Let us define $\mathcal{P}_k$ as the family of probabilities with mean μ_k and covariance Σ_k :

$$\mathcal{P}_k \doteq \{P : \mathbb{E}_P[X] = \mu_k, \mathbb{E}_P[(X - \mu_k)(X - \mu_k)^T] = \Sigma_k\}. \quad (38)$$

This is a closed convex set of probabilities and a linear family in the sense of (37), as it is defined by linear equality constraints on the sufficient statistics ($P \mapsto \mathbb{E}_P[X]$ and $P \mapsto \mathbb{E}_P[XX^T]$ are linear functionals of P), so the Pythagorean equality holds exactly.

The I-projection of Amari [1983] of $p_0 = \mathbb{P}_{r,X|Y=k}^{\text{LDA}} \notin \mathcal{P}_k$ onto $\mathcal{P}_k$ is indeed $\mathbb{P}_{r,X|Y=k}^{\text{QDA}}$, i.e. $\mathcal{N}(\mu_k, \Sigma_k)$. We write $\text{KL}(p||p_0) = \mathbb{E}_p[-\log p_0(X)] - H(p)$. Since p_0 is a Gaussian distribution, $-\log p_0(X)$ is quadratic in X whose expectation depends only on the statistics (μ_k, Σ_k) . In other words, this cross-entropy term is constant over $\mathcal{P}_k$. Minimizing $\text{KL}(p||p_0)$ then amounts to maximizing the entropy $H(p)$ over $\mathcal{P}_k$, which is achieved by $\mathcal{N}(\mu_k, \Sigma_k)$ Cover and Thomas [2006].

$$p^\star = \arg \min_{p \in \mathcal{P}_k} \text{KL}(p||\mathbb{P}_{r,X|Y=k}^{\text{LDA}}) = \mathbb{P}_{r,X|Y=k}^{\text{QDA}}.$$

Assuming that $\mathbb{P}_{r,X|Y=k} \in \mathcal{P}_k$, the equality holds exactly:

$$\text{KL}_{X|Y=k}(\mathbb{P}_r||\mathbb{P}^{\text{LDA}}) = \text{KL}_{X|Y=k}(\mathbb{P}_r||\mathbb{P}^{\text{QDA}}) + \text{KL}_{X|Y=k}(\mathbb{P}^{\text{QDA}}||\mathbb{P}^{\text{LDA}}).$$

Lemma 3. *The KL between two Gaussians with the same mean μ_k and different covariances Σ_k and Σ respectively:*

$$\text{KL}(\mathcal{N}(\mu_k, \Sigma_k)||\mathcal{N}(\mu_k, \Sigma)) = \frac{1}{2} \left(\log \frac{|\Sigma|}{|\Sigma_k|} + \text{tr}(\Sigma^{-1}\Sigma_k) - d \right).$$

Finally, we link the above lemma to Prop. 2 with the following result:

Proposition 4. *The explicit cost of the homoscedastic assumption, i.e. LDA proxy, on the bound of Prop. 2.*

$$\text{KL}_{X|Y}(\mathbb{P}_r||\mathbb{P}_r^{\text{LDA}}) = \text{KL}_{X|Y}(\mathbb{P}_r||\mathbb{P}_r^{\text{QDA}}) + \frac{1}{2} \sum_{k=1}^C \mathbb{P}_r(k) \left(\log \frac{|\Sigma|}{|\Sigma_k|} + \text{tr}(\Sigma^{-1}\Sigma_k) - d \right).$$

Then, an unlearner knows exactly the cost to pay in term of divergence when wanting to reduce the computational cost of various inverse covariances.

F Exact Unlearning Signal with empirical distribution

A proxy distribution that can be thought of is the empirical measure. What happens if we model the data structure with indicator function over the labeled data $(x, y(x))$?

We derive a closed form $\Delta M(x)$ (2) for this proxy and discuss its consequences for the unlearned model $\tilde{f}_\eta = f_\theta + \eta \Delta M$ (39) and for the KL distillation (3). We use the framework of mixture models described in Sect. 3; in particular the mixture $P(x|y) = \pi_r(y)P_r(x|y) + \pi_f(y)P_f(x|y)$ and posteriors (1).

$$\tilde{f}_\eta(x) \doteq f_\theta(x) + \eta \Delta M(x), \quad \Delta M(x) = \log P_r(\cdot|x) - \log P(\cdot|x), \quad 0 \leq \eta \leq 1. \quad (39)$$

We propose to take both proxies are empirical measures on their respective support:

$$P_r(x|y) = \frac{1}{|\mathcal{D}_r(y)|} \mathbf{1}\{(x, y) \in \mathcal{D}_r\},$$

and likewise for P_f on $\mathcal{D}_f$. P becomes a mixture of empirical measures on $\mathcal{D}$:

$$P(x|y) = \frac{\pi_r(y)}{|\mathcal{D}_r(y)|} \mathbf{1}\{(x, y) \in \mathcal{D}_r\} + \frac{\pi_f(y)}{|\mathcal{D}_f(y)|} \mathbf{1}\{(x, y) \in \mathcal{D}_f\}.$$

Since $\frac{\pi_r(y)}{|\mathcal{D}_r(y)|} = \frac{|\mathcal{D}_r(y)|}{|\mathcal{D}(y)||\mathcal{D}_r(y)|} = \frac{1}{|\mathcal{D}(y)|}$, we simply recover:

$$P(x|y) = \frac{1}{|\mathcal{D}(y)|} \mathbf{1}\{(x, y) \in \mathcal{D}\},$$

which is the initial empirical measure. This is the same as doing *no* mixture.

For any $(x, y) \in \mathcal{D}_r$, we have $P_r(y) = \mathbb{P}_r(y)$ and $P_r(x) = \sum_c \mathbb{P}_r(c)P_r(x|c) = \mathbb{P}_r(y)P_r(x|y)$. The Bayes inversion (1) leaves $P_r(y|x) = P_r(x|y)\mathbb{P}_r(y)/P_r(x) = 1$. This also holds for P .

$$P_r(y|x) = \mathbf{1}\{(x, y) \in \mathcal{D}_r\}, \quad P(y|x) = \mathbf{1}\{(x, y) \in \mathcal{D}\}$$

Substituting in (39) gives

$$\Delta M(x)_y = \log \frac{\mathbf{1}\{(x, y) \in \mathcal{D}_r\}}{\mathbf{1}\{(x, y) \in \mathcal{D}_r\} + \mathbf{1}\{(x, y) \in \mathcal{D}_f\}}.$$

Since $\mathcal{D}_r \cap \mathcal{D}_f = \emptyset$, we obtain the closed form

$$\Delta M(x)_y = \log \mathbf{1}\{(x, y) \in \mathcal{D}_r\}. \quad (40)$$

This gives an unlearn model $\tilde{f}$ which completely annihilates the (x, y) data (logit set to $-\infty$) unless it belongs to the retain set where the logit is left unchanged.

No more dependency on η For any $\eta > 0$, the $-\infty$ entries dominates the softmax of $\tilde{f}_\eta$ uniformly, leaving the truncated probit in the limit:

$$\forall x \notin \mathcal{D}_r, \tilde{p}_\eta(x)_c = \begin{cases} 0 & c = y(x), \\ p_\theta(x)_c / (1 - p_\theta(x)_{y(x)}) & c \neq y(x). \end{cases} \quad (41)$$

The truncated probit (41) carries zero mass at y on the forget set. On the other hand, for all $x \in \mathcal{D}_r$, $\tilde{p}_\eta(x) = p_\theta(x)$. The unlearned probit is η -independent. The line search of Sect. 3.2 is therefore degenerated.

Consequence on the distillation pipeline As it is, this correction makes little sense in practice, *i.e.* for unseen testing data. Yet, the KL distillation (3) over $\mathcal{D}$ remains well-posed because $\text{KL}(\tilde{p} \parallel \text{softmax}(f_{\theta+\delta\theta}))$ uses $\tilde{p}$ as the left argument and $0 \cdot \log 0 = 0$. The KL distillation drives the unlearned NN to get closer to the truncated probit (41), but *not* exactly equal to. This makes a strong decrease of the probit related to $y(x)$ for the forget data.

Empirical exact unlearning in 2C dimensions The C-dimensional truncation (41) is too aggressive in the *Subclass* scenario. We lift this truncation from labels $y \in \llbracket C \rrbracket$ to 2C-dimensions labels $(y, s) \in \llbracket C \rrbracket \times \{r, f\}$. This amounts to canceling the mass probability of $(y(x), f)$ for $x \in \mathcal{D}_f$ and redistributing this quantity to the remaining cells (y, s) . Each s -row needs a renormalization constant, equal to the probability $\pi_s(x)$ of a given sample x to be in the retain ($s = r$) or the forget set ($s = f$). The simplest inference uses a feature agnostic assumption: $P(s|x, y) \simeq P(s|y) = \pi_s(y) = |\mathcal{D}_s(y)|/|\mathcal{D}(y)|$:

$$\pi_s(x) = \sum_y P(s|x, y) P(y|x) \simeq \hat{\pi}_s(x) \doteq \sum_y P(y|x) \pi_s(y).$$

Since $P(y|x) = \mathbf{1}\{(x, y) \in \mathcal{D}\} = \mathbf{1}\{y = y(x)\}$, then $\hat{\pi}_s(x) = \pi_s(y(x))$. One can retrieve the dimension C classifier:

$$\tilde{p}(x) \doteq \hat{\pi}_r(x) p_\theta(x) + \hat{\pi}_f(x) \tilde{p}_f(x),$$

with $\tilde{p}_f(x)$ defined as the previous the C dimension unlearned model (41). In the end, we have a feature-agnostic unlearned model using only the empirical distributions. Again, it makes little sense in practice, *i.e.* for unseen testing data. The KL distillation drives the unlearned NN to get closer to this target. This results in a slight decrease of the probit related to $y(x)$ for the forget data. This approximation makes experimentally a strong unlearning method for the *Subclass* scenario.

G Hypothesis Test and KL divergence

This section proposes an alternative to the evaluation via a Member Inference Attack.

Imagine that an attacker has access to a classifier in a black-box. He knows that this black box contains either the ideal model retrained from scratch $\mathbb{P}_r$ or the unlearned NN $p_{\tilde{\theta}}$. He queries the back box with some random data X and observe the output $\hat{Y}$, the predicted label. Of note, this is a worst case scenario for the attacker. MIAs in the literature agree on a more favorable scenario where the attack has access to the probit or logit vector.

This section makes the connection between KL_f (or KL_t), as reported in the experimental results, and the number of queries necessary to identify the classifier in the black-box with a given reliability.

For a given input x , we have the following hypothesis:

$$\mathcal{H}_0: \hat{Y} \sim \mathbb{P}_{rY|x}$$

$$\mathcal{H}_1: \hat{Y} \sim p_{\tilde{\theta}Y|x}$$

The attacker's test outputs a binary decision $D \in \{0, 1\}$, which has the following Bernoulli distribution:

$\mathcal{H}_0$: The test makes a *false positive* whenever $D = 1$. Let us denote the probability of false positive by α . Then, $D \sim \mathcal{B}(\alpha)$.

$\mathcal{H}_1$: The test makes a *false negative* whenever $D = 0$. Let us denote the probability of false negative by β . Then, $D \sim \mathcal{B}(1 - \beta)$.

Suppose that the attacker design the test for achieving an Equal Error Rate, *i.e.* $\alpha = \beta$, since the error rates α and β matters equally.

The data processing theorem states that any processing of the data $\hat{Y}$ reduces the KL divergence. The attacker's test is a processing, therefore:

$$\begin{aligned} KL_{Y|x}(\mathbb{P}_r \| p_{\tilde{\theta}}) &\geq KL(\mathcal{B}(\alpha) \| \mathcal{B}(1 - \beta)) = (1 - \alpha) \log\left(\frac{1 - \alpha}{\beta}\right) + \alpha \log\left(\frac{\alpha}{1 - \beta}\right) \\ &\stackrel{(\alpha=\beta)}{=} (1 - 2\alpha) \log\left(\frac{1 - \alpha}{\alpha}\right). \end{aligned}$$

This inequality is at the root of the Stein lemma.

Querying $N > 1$ inputs brings more information. Provided they are independent (assumption A1), their KL divergences sum up. This sum can be written $N \cdot \mathbb{E}KL_{Y|x}(\mathbb{P}_r \| p_{\tilde{\theta}})$ where the expectation

is indeed an average over the queries as reported experimentally in Sect. 5. At the end, we get a necessary number of queries to reach a reliability given by $1 - \alpha$:

$$N \geq \frac{(1 - 2\alpha) \log(1 - \alpha/\alpha)}{\mathbb{E} \text{KL}_{Y|X}(\mathbb{P}_r \| \mathbb{P}_{\hat{\theta}})}.$$

This number comes with two flavors. The attacker may query test data (resp. forget data) and we plug KL_t (resp. KL_f) in the formula. For instance, for $\alpha = 1/1\,000$, $N_t = 138$ for $\text{KL}_t = 0.05$, $N_f = 69$ for $\text{KL} = 0.1$, which are typical values reported in Sect. 5. The good question is to compare N_f with the actual number of forget data: if $|\mathcal{D}_f| < N_f$, then the attacker is unable to distinguish the two classifiers with forget data, and even if he knows all of them.

H Handling large NN

This appendix shows that scaling our framework to networks whose hidden representations are too high-dimensional for a direct LDA/QDA fit requires *no theoretical change*. It is purely a question of choice of the proxies.

Composed proxy The whole machinery of Sect. 3 only requires to model the input distribution so that $\Delta M(x) = M_r(x) - M(x)$ can be computed in closed form. Up to now we instantiated this proxy directly on x via LDA or QDA. When x lives in a space too large for a class-conditional Gaussian fit, we simply compose:

$$P(x|y) \doteq P_0(F(x)|y), \quad F: \mathcal{X} \rightarrow \mathbb{R}^k,$$

where P_0 is the standard LDA or QDA proxy on the space $F(\mathcal{X})$ and F is any fixed injection from the high-dimensional input space onto a low-dimensional target. The retain proxy P_r is built identically with the same F . Plugging (H) into the framework yields a logit shift $\Delta M(x) = \Delta M_0(F(x))$ that is computed in $\mathbb{R}^k$. All the closed-form properties of discriminant analysis are preserved. Assumptions and the bounds of Sect. 3 carry over verbatim. Any F that yields a low-dimensional representation on which LDA/QDA is accurate is admissible.

Practical instance for ResNet-18 The specific F we use in the ResNet-18 experiments is anecdotal — it is one convenient choice among many. We take

$$F(x) = \Omega \phi^{(3)}(x), \tag{42}$$

where $\phi^{(3)}$ is the activation just after the third residual stage ($\dim \sim 8 \times 10^3$ at our resolution) and $\Omega \in \mathbb{R}^{k \times d}$ ($k=512$) is a deterministic semi-orthogonal Johnson–Lindenstrauss sketch [Johnson and Lindenstrauss, 1984, Dasgupta and Gupta, 2003] obtained as the orthogonal factor of a Gaussian matrix via QR decomposition, with a fixed seed shared across all proxy variants. The hook depth ($\phi^{(3)}$) is the standard mid-level/pre-classification choice: earlier stages have too small a receptive field, later ones already encode logits and would make the proxy redundant with f_θ .

F enters only the proxy and enables recovering approximately Gaussian per-class features, as the semi-orthogonal projection of the high-dimensional activations induces a sub-Gaussian design on each class conditional [Dereziński, 2023, Vershynin, 2018]. The MSE finalizer regresses the per-sample target $\Delta(x) = \eta_{\max} \Delta M_0(F(x))$ on the network’s own penultimate representation $\phi(x)$ by closed-form least squares. The alternative finalizer, the KL distillation, distils the full logit target into the network by backpropagation. The proxy and the finalizer may live in different representations, exactly as in the DINOv2 pipeline.

Scalability This framework scales to arbitrary backbones at fixed cost in k because the proxy fit is performed in $\mathbb{R}^k$ regardless of the input dimension. Operator F is the only choice that needs revisiting for a new architecture: any low-dimensional representation on which LDA/QDA is reasonable (an intermediate activation, a learned embedding, a sketch of any of these) is a valid drop-in. One has to construct F to make the good proxy assumptions hold.

I Experiments — supplementary material

I.1 Setup details

Protocol. Datasets: CIFAR-10 / CIFAR-100 [Krizhevsky, 2009] under licence MIT (50k / 10k torchvision splits). DINOv2-Small features ($d=384$) [Oquab et al., 2024] are pre-extracted once; ResNet-18 takes raw 32×32 images upsampled to 64×64 . CIFAR-100’s 20 superclasses define the *Subclass* head. Frozen DINOv2 heads are trained from scratch with Adam at $\text{lr}=10^{-3}$ for 50 epochs, batch 512; ResNet-18 is fine-tuned from ImageNet for 20 epochs at $\text{lr}=10^{-4}$, batch 128. Iterative methods share the same recipe per backbone (multiplicative decay $\gamma=0.95$, 20 epochs); for each method we record every epoch and select the best by $\min \text{KL}_f$ on the training data, uniform across methods. Five seeds $\{42, 0, 1, 2, 3\}$, five class / subclass IDs (CIFAR-10: $\{0, 2, 4, 6, 8\}$, CIFAR-100: $\{0, 20, 40, 60, 80\}$), $n_f \in \{1, 50, 500, 5000\}$ for *Random*.

Hardware. A cluster of GPUs V100 / A100 nodes; cells were not pinned to a single machine, so we report *normalised* wall-clock RTE/RTE_{RETRAIN} (per-seed, same node). We made the computation across a cluster of GPUs. Total compute for the full grid is on the order of 200 GPU-hours. Code release upon publication; every table / figure regenerates with one command from the released raw JSON results.

I.2 Auxiliary ablations

I.2.1 Structure of the released results/ folder

To enable an exhaustive replication of every tables in the paper, the full set of raw results is released as a JSON tree in the supplementary file. This represents 2 datasets $\times$ 4 architectures $\times$ scenarios $(5 + 5 + 4) = 112$ variants with the raw per-seed outputs of every run. Layout:

```
results/
  {dataset}/
    {arch_kind}/
      {scenario}_{arch}_raw.json
      # scenario : class | subclass | random
      # arch      : linear | mlp1 | mlp2 | resnet18
```

Each JSON file is keyed first by the architecture short name, then either flat $\{\text{sub_key}: [\text{seed_entry}, \dots]\}$ (for CLASS, RANDOM) or wrapped as $\{\text{"meta": } \dots, \text{"results": } \{\text{sub_key}: [\dots]\}\}$ (for SUBCLASS). Sub-key conventions:

- *Class* — class index of the forgotten class. CIFAR-10: $\{0, 2, 4, 6, 8\}$, CIFAR-100: $\{0, 20, 40, 60, 80\}$.
- *Subclass* — fine class index inside its superclass; same sets as above.
- *Random* — number of forgotten samples $n_f \in \{1, 50, 500, 5000\}$.

A *seed_entry* is a dict whose top-level keys are method names. Per method we store a *best* block (best epoch by KL_f on the train set), an *epochs* list (one entry per epoch with all metrics), and (for our variants) a *target* block reporting the proxy metrics ($\eta_{\max}$, KL_t , KL_f , $\text{KL}_{\text{NET} \rightarrow \text{PROXY}}$ before/after, ...). The reference rows initial ($\mathbb{P}$) and retrained ($\mathbb{P}_r$) are stored at the top level of each seed entry.

I.2.2 Ablation tables (non-exhaustive)

The presented results are best KL over epochs: it favors the SOTA baselines, it has to be put in contrast with the last epoch KL pointing the stability of our method. As suggested in Fig. 2 SOTA baselines are unstable through time or show inefficacy in terms of KL when there is something to unlearn: if the KL of the initial model from the retrained is high.

Across forgetting scenarios (CIFAR-10, DINOv2 + MLP-1). Reproduces the main table on *Subclass* and adds the *Class* and *Random* companions at their default sub-keys.

Table 5: Benchmark on CIFAR10 / dinov2-mlp1, scenario *CLASS* (airplane). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_{θ}	1.5 ± 0.02	–	16 ± 0.19	96.9 ± 0.1	100.0 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	87.3 ± 0.1	0.0 ± 0.0	100 (ref)
LDA	0.13 ± 0.01	0.14 ± 0.01	1.2 ± 0.12	87.2 ± 0.1	0.0 ± 0.0	4.6 ± 0.27
LDA-Mix	0.07 ± 0.01	0.07 ± 0.01	0.52 ± 0.08	87.2 ± 0.1	0.1 ± 0.1	5.2 ± 0.49
LDA-2C	0.07 ± 0.01	0.07 ± 0.01	0.58 ± 0.09	87.2 ± 0.1	0.0 ± 0.0	4.7 ± 0.22
DIR	0.07 ± 0.01	0.07 ± 0.01	0.58 ± 0.09	87.2 ± 0.1	0.0 ± 0.0	4.8 ± 0.35
DIR-2C	0.07 ± 0.01	0.07 ± 0.01	0.58 ± 0.11	87.1 ± 0.1	0.0 ± 0.0	4.7 ± 0.31
SCRUB	0.23 ± 0.11	1.6 ± 0.36	2.1 ± 1.1	87.4 ± 0.3	4.8 ± 2.9	6.4 ± 0.34
SalUn	0.21 ± 0.00	0.21 ± 0.00	1.5 ± 0.04	86.3 ± 0.2	0.0 ± 0.0	4.4 ± 0.14
RL+FT	0.18 ± 0.00	0.20 ± 0.00	1.5 ± 0.05	87.0 ± 0.1	0.0 ± 0.0	4.7 ± 0.43
GA+FT	0.68 ± 0.25	0.68 ± 0.25	7.0 ± 2.7	87.2 ± 0.1	0.0 ± 0.0	18 ± 1.1
FT	0.57 ± 0.05	0.57 ± 0.05	5.7 ± 0.55	94.5 ± 0.3	74.8 ± 2.7	17 ± 0.96
GA	1.4 ± 0.17	876 ± 43	8.2 ± 1.5	79.1 ± 0.9	0.0 ± 0.0	2.1 ± 0.17

Table 6: Benchmark on CIFAR10 / dinov2-linear, scenario *SUBCLASS* (airplane). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_{θ}	0.39 ± 0.01	–	3.8 ± 0.11	99.5 ± 0.0	99.2 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	95.9 ± 0.1	62.0 ± 1.0	100 (ref)
LDA	0.05 ± 0.00	0.05 ± 0.00	0.40 ± 0.02	95.1 ± 0.0	53.6 ± 0.1	5.1 ± 0.79
LDA-Mix	0.15 ± 0.01	0.15 ± 0.00	1.5 ± 0.06	98.9 ± 0.0	92.3 ± 0.2	5.3 ± 0.31
LDA-2C	0.04 ± 0.00	0.04 ± 0.00	0.35 ± 0.01	96.3 ± 0.0	63.3 ± 0.2	4.8 ± 0.51
DIR	0.11 ± 0.00	0.46 ± 0.01	0.63 ± 0.02	92.8 ± 0.2	40.3 ± 1.9	5.0 ± 0.76
DIR-2C	0.06 ± 0.00	0.06 ± 0.00	0.44 ± 0.02	96.2 ± 0.0	66.0 ± 0.5	5.1 ± 0.74
SCRUB	0.05 ± 0.00	0.06 ± 0.00	0.52 ± 0.02	96.0 ± 0.3	64.1 ± 2.4	7.2 ± 1.2
SalUn	0.13 ± 0.00	0.55 ± 0.01	0.60 ± 0.02	93.2 ± 0.2	51.4 ± 1.3	5.0 ± 0.74
RL+FT	0.11 ± 0.00	0.48 ± 0.02	0.63 ± 0.02	92.8 ± 0.2	40.3 ± 2.0	4.6 ± 0.34
GA+FT	0.06 ± 0.01	0.06 ± 0.01	0.62 ± 0.05	97.9 ± 0.0	82.6 ± 0.3	19 ± 2.1
FT	0.04 ± 0.00	0.04 ± 0.01	0.36 ± 0.04	97.3 ± 0.1	77.6 ± 0.9	18 ± 2.6
GA	0.14 ± 0.01	19 ± 0.11	0.68 ± 0.02	93.4 ± 0.2	57.0 ± 1.5	1.9 ± 0.21

Across architectures (CIFAR-10, scenario *Subclass*, sub-key 0). Same scenario as the main table, but the head varies from a single linear layer to a 2-hidden-layer MLP and finally to a fully fine-tuned ResNet-18.

Across datasets (DINOv2 + MLP-1, scenario *Subclass*, sub-key 0). The CIFAR-100 counterpart of the main table.

Across sub-keys (CIFAR-10, DINOv2 + MLP-1, scenario *Subclass*). We pick an available sub-keys to assess sensitivity to the identity of the forgotten subclass.

Across forget-set size (CIFAR-10, DINOv2 + MLP-1, scenario *Random*). Sweeping $n_f \in \{1, 500, 5000\}$ around the default $n_f = 50$.

The qualitative ranking observed on the headline cell — LDA-2C and DIR-2C are competitive with the strongest SOTA on KL-based metrics while paying a tiny RTE is preserved across every architecture, dataset and sub-key reported here. The quantitative spread is small enough that no scenario inverts the ordering between our methods and the closest SOTA baseline.

Short Ablation across Scenarios and Architectures

Table 7: Benchmark on CIFAR10 / dinov2-mlp2, scenario *SUBCLASS* (airplane). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.48 ± 0.02	–	4.8 ± 0.20	99.7 ± 0.0	100.0 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	95.7 ± 0.2	60.3 ± 1.4	100 (ref)
LDA	0.05 ± 0.01	0.07 ± 0.00	0.49 ± 0.07	93.9 ± 0.3	42.6 ± 2.8	5.1 ± 0.06
LDA-Mix	0.06 ± 0.00	0.08 ± 0.00	0.63 ± 0.04	98.5 ± 0.0	88.3 ± 0.6	5.4 ± 0.67
LDA-2C	0.04 ± 0.00	0.05 ± 0.00	0.40 ± 0.03	94.7 ± 0.6	47.1 ± 5.7	5.0 ± 0.08
DIR	0.47 ± 0.02	0.62 ± 0.03	4.9 ± 0.14	89.9 ± 0.2	2.6 ± 1.3	5.0 ± 0.05
DIR-2C	0.05 ± 0.00	0.05 ± 0.00	0.50 ± 0.02	99.5 ± 0.1	99.2 ± 0.3	5.0 ± 0.06
SCRUB	0.11 ± 0.02	3.3 ± 0.19	1.1 ± 0.17	96.2 ± 1.2	65.8 ± 12.1	6.7 ± 0.10
SalUn	0.77 ± 0.05	0.81 ± 0.06	8.1 ± 0.52	89.6 ± 0.1	0.3 ± 0.1	4.7 ± 0.12
RL+FT	0.48 ± 0.04	0.63 ± 0.05	5.0 ± 0.43	90.0 ± 0.2	3.3 ± 0.7	4.4 ± 0.05
GA+FT	0.82 ± 0.30	0.84 ± 0.29	9.2 ± 3.5	91.9 ± 0.3	19.5 ± 2.4	19 ± 0.32
FT	0.26 ± 0.03	0.37 ± 0.06	2.7 ± 0.32	99.2 ± 0.3	95.0 ± 2.7	19 ± 0.70
GA	11 ± 3.6	24650 ± 2051	28 ± 8.8	60.1 ± 0.1	0.0 ± 0.0	2.2 ± 0.07

Table 8: Benchmark on CIFAR10 / resnet18, scenario *SUBCLASS* (airplane). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.51 ± 0.03	–	5.3 ± 0.25	98.0 ± 0.1	100.0 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	93.7 ± 0.1	49.1 ± 0.6	100 (ref)
LDA	0.21 ± 0.03	0.21 ± 0.02	2.0 ± 0.24	97.0 ± 0.2	98.9 ± 1.1	21 ± 0.85
LDA-Mix	0.20 ± 0.02	0.21 ± 0.01	2.0 ± 0.17	97.1 ± 0.3	98.6 ± 1.1	21 ± 0.85
LDA-2C	0.20 ± 0.01	0.21 ± 0.01	2.0 ± 0.16	97.3 ± 0.2	99.2 ± 0.6	21 ± 0.94
DIR	0.22 ± 0.03	0.29 ± 0.05	0.79 ± 0.06	92.4 ± 1.2	79.8 ± 5.0	18 ± 0.79
DIR-2C	0.16 ± 0.00	0.18 ± 0.01	1.5 ± 0.19	96.8 ± 0.8	97.2 ± 2.6	18 ± 0.76
SCRUB	0.41 ± 0.03	0.45 ± 0.05	3.6 ± 0.23	95.9 ± 0.2	92.4 ± 2.9	19 ± 0.83
SalUn	0.28 ± 0.03	0.33 ± 0.05	0.60 ± 0.02	86.2 ± 2.6	52.0 ± 10.9	15 ± 0.60
RL+FT	0.22 ± 0.07	0.29 ± 0.05	0.81 ± 0.04	92.2 ± 2.6	79.8 ± 10.6	14 ± 0.57
GA+FT	0.46 ± 0.03	0.60 ± 0.03	5.0 ± 0.18	97.7 ± 0.2	99.7 ± 0.3	65 ± 2.8
FT	0.58 ± 0.04	0.73 ± 0.07	6.1 ± 0.13	97.8 ± 0.2	99.7 ± 0.6	64 ± 2.6
GA	1.3 ± 0.28	2.7 ± 0.41	2.9 ± 0.13	82.9 ± 2.8	45.5 ± 6.5	7.0 ± 0.25

Table 9: Benchmark on CIFAR100 / dinov2-mlp1, scenario *SUBCLASS* (apple). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.08 ± 0.00	–	0.35 ± 0.07	92.4 ± 0.0	100.0 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	92.3 ± 0.1	96.2 ± 0.7	100 (ref)
LDA	0.11 ± 0.01	0.09 ± 0.00	0.32 ± 0.07	91.9 ± 0.2	100.0 ± 0.0	1.2 ± 0.03
LDA-Mix	0.09 ± 0.00	0.09 ± 0.00	0.12 ± 0.02	92.2 ± 0.0	94.9 ± 2.3	1.5 ± 0.43
LDA-2C	0.10 ± 0.01	0.09 ± 0.00	0.32 ± 0.07	92.1 ± 0.2	100.0 ± 0.0	1.2 ± 0.03
DIR	0.09 ± 0.00	0.15 ± 0.01	0.31 ± 0.07	92.2 ± 0.1	84.5 ± 3.9	1.1 ± 0.05
DIR-2C	0.09 ± 0.00	0.09 ± 0.00	0.18 ± 0.03	92.2 ± 0.1	98.7 ± 0.4	1.2 ± 0.05
SCRUB	0.09 ± 0.00	0.09 ± 0.00	0.28 ± 0.12	92.3 ± 0.1	98.2 ± 2.3	0.63 ± 0.03
SalUn	0.09 ± 0.00	0.52 ± 0.03	0.10 ± 0.02	92.3 ± 0.0	96.7 ± 0.2	0.45 ± 0.04
RL+FT	0.09 ± 0.00	0.26 ± 0.01	0.16 ± 0.02	92.2 ± 0.0	93.2 ± 1.3	0.60 ± 0.05
GA+FT	0.13 ± 0.01	0.13 ± 0.00	0.19 ± 0.06	91.6 ± 0.2	98.8 ± 0.4	17 ± 0.86
FT	0.12 ± 0.00	0.13 ± 0.00	0.29 ± 0.05	91.9 ± 0.1	100.0 ± 0.1	17 ± 0.64
GA	0.09 ± 0.00	4.2 ± 0.14	0.13 ± 0.02	92.2 ± 0.1	93.3 ± 0.8	0.21 ± 0.01

Table 10: Benchmark on CIFAR100 / resnet18, scenario *SUBCLASS* (apple). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.90 $\pm$ 0.02	–	1.4 $\pm$ 0.19	77.0 $\pm$ 0.3	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	77.0 $\pm$ 0.4	86.4 $\pm$ 1.5	100 (ref)
LDA	0.93 $\pm$ 0.01	0.93 $\pm$ 0.01	0.81 $\pm$ 0.06	72.1 $\pm$ 0.2	89.3 $\pm$ 1.3	7.7 $\pm$ 0.09
LDA-Mix	0.93 $\pm$ 0.01	0.93 $\pm$ 0.01	0.80 $\pm$ 0.06	72.2 $\pm$ 0.2	89.4 $\pm$ 1.2	7.7 $\pm$ 0.09
LDA-2C	0.92 $\pm$ 0.01	0.92 $\pm$ 0.01	0.80 $\pm$ 0.06	72.3 $\pm$ 0.2	89.2 $\pm$ 1.0	7.6 $\pm$ 0.12
DIR	1.1 $\pm$ 0.01	0.95 $\pm$ 0.01	0.87 $\pm$ 0.09	74.5 $\pm$ 0.2	92.8 $\pm$ 1.6	5.2 $\pm$ 0.14
DIR-2C	0.95 $\pm$ 0.06	0.92 $\pm$ 0.01	0.81 $\pm$ 0.06	72.7 $\pm$ 0.9	90.2 $\pm$ 2.5	5.2 $\pm$ 0.10
SCRUB	1.2 $\pm$ 0.05	1.4 $\pm$ 0.04	0.93 $\pm$ 0.15	74.3 $\pm$ 0.3	93.3 $\pm$ 1.6	1.8 $\pm$ 0.03
SalUn	1.1 $\pm$ 0.01	1.0 $\pm$ 0.02	0.88 $\pm$ 0.10	73.8 $\pm$ 0.3	92.6 $\pm$ 1.4	1.4 $\pm$ 0.03
RL+FT	1.1 $\pm$ 0.01	1.1 $\pm$ 0.02	0.87 $\pm$ 0.10	74.0 $\pm$ 0.2	92.4 $\pm$ 1.3	0.67 $\pm$ 0.04
GA+FT	0.88 $\pm$ 0.02	0.97 $\pm$ 0.02	1.4 $\pm$ 0.21	77.1 $\pm$ 0.2	100.0 $\pm$ 0.0	84 $\pm$ 0.78
FT	0.99 $\pm$ 0.03	1.2 $\pm$ 0.03	1.6 $\pm$ 0.20	76.9 $\pm$ 0.3	100.0 $\pm$ 0.0	84 $\pm$ 0.71
GA	1.2 $\pm$ 0.05	2.5 $\pm$ 0.11	0.94 $\pm$ 0.15	74.1 $\pm$ 0.4	92.4 $\pm$ 3.1	0.39 $\pm$ 0.01

Table 11: Benchmark on CIFAR10 / dinov2-mlp1, scenario *SUBCLASS* (deer). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.02 $\pm$ 0.00	–	0.22 $\pm$ 0.02	99.7 $\pm$ 0.0	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	99.6 $\pm$ 0.0	98.0 $\pm$ 0.1	100 (ref)
LDA	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.18 $\pm$ 0.02	99.6 $\pm$ 0.0	100.0 $\pm$ 0.0	4.8 $\pm$ 0.15
LDA-Mix	0.01 $\pm$ 0.00	0.01 $\pm$ 0.00	0.11 $\pm$ 0.02	99.7 $\pm$ 0.1	100.0 $\pm$ 0.0	5.1 $\pm$ 0.41
LDA-2C	0.01 $\pm$ 0.00	0.01 $\pm$ 0.00	0.08 $\pm$ 0.01	99.7 $\pm$ 0.0	99.9 $\pm$ 0.0	4.7 $\pm$ 0.28
DIR	0.65 $\pm$ 0.05	0.75 $\pm$ 0.03	6.6 $\pm$ 0.52	90.5 $\pm$ 0.3	10.3 $\pm$ 3.8	4.6 $\pm$ 0.19
DIR-2C	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.15 $\pm$ 0.01	99.6 $\pm$ 0.1	99.2 $\pm$ 0.7	4.7 $\pm$ 0.28
SCRUB	0.01 $\pm$ 0.00	2.3 $\pm$ 0.05	0.11 $\pm$ 0.02	99.6 $\pm$ 0.2	99.0 $\pm$ 2.0	6.2 $\pm$ 0.44
SalUn	1.0 $\pm$ 0.03	1.0 $\pm$ 0.04	10 $\pm$ 0.28	88.4 $\pm$ 0.1	1.0 $\pm$ 0.1	4.4 $\pm$ 0.34
RL+FT	0.66 $\pm$ 0.05	0.76 $\pm$ 0.02	6.7 $\pm$ 0.42	90.5 $\pm$ 0.2	10.4 $\pm$ 3.7	4.1 $\pm$ 0.20
GA+FT	0.16 $\pm$ 0.04	0.16 $\pm$ 0.04	1.7 $\pm$ 0.47	98.0 $\pm$ 0.2	81.4 $\pm$ 2.5	18 $\pm$ 0.95
FT	0.01 $\pm$ 0.00	0.01 $\pm$ 0.00	0.13 $\pm$ 0.03	99.7 $\pm$ 0.1	99.8 $\pm$ 0.1	17 $\pm$ 0.80
GA	13 $\pm$ 0.59	1338 $\pm$ 49	46 $\pm$ 1.9	45.5 $\pm$ 0.4	0.0 $\pm$ 0.0	2.0 $\pm$ 0.08

Table 12: Benchmark on CIFAR10 / dinov2-mlp1, scenario *SUBCLASS* (frog). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.03 $\pm$ 0.00	–	0.31 $\pm$ 0.01	99.7 $\pm$ 0.0	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	99.4 $\pm$ 0.0	97.1 $\pm$ 0.1	100 (ref)
LDA	0.03 $\pm$ 0.00	0.03 $\pm$ 0.00	0.25 $\pm$ 0.02	99.6 $\pm$ 0.1	100.0 $\pm$ 0.0	5.0 $\pm$ 0.07
LDA-Mix	0.03 $\pm$ 0.00	0.03 $\pm$ 0.00	0.24 $\pm$ 0.02	99.6 $\pm$ 0.0	100.0 $\pm$ 0.0	5.5 $\pm$ 0.69
LDA-2C	0.03 $\pm$ 0.00	0.03 $\pm$ 0.00	0.24 $\pm$ 0.03	99.6 $\pm$ 0.1	99.9 $\pm$ 0.1	5.0 $\pm$ 0.05
DIR	1.0 $\pm$ 0.10	1.3 $\pm$ 0.05	10 $\pm$ 0.98	89.8 $\pm$ 0.2	6.7 $\pm$ 1.0	5.0 $\pm$ 0.08
DIR-2C	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.17 $\pm$ 0.01	99.6 $\pm$ 0.0	99.9 $\pm$ 0.0	5.0 $\pm$ 0.09
SCRUB	0.02 $\pm$ 0.00	4.7 $\pm$ 0.10	0.12 $\pm$ 0.04	99.2 $\pm$ 0.3	95.9 $\pm$ 3.3	6.6 $\pm$ 0.13
SalUn	1.5 $\pm$ 0.06	1.5 $\pm$ 0.05	14 $\pm$ 0.56	89.0 $\pm$ 0.1	0.2 $\pm$ 0.1	4.5 $\pm$ 0.11
RL+FT	1.0 $\pm$ 0.04	1.3 $\pm$ 0.06	10 $\pm$ 0.36	89.7 $\pm$ 0.1	6.0 $\pm$ 0.5	4.4 $\pm$ 0.08
GA+FT	0.20 $\pm$ 0.05	0.20 $\pm$ 0.05	2.1 $\pm$ 0.50	97.0 $\pm$ 0.3	71.8 $\pm$ 2.8	19 $\pm$ 0.33
FT	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.17 $\pm$ 0.02	99.6 $\pm$ 0.0	99.7 $\pm$ 0.2	19 $\pm$ 0.39
GA	11 $\pm$ 0.36	1230 $\pm$ 35	43 $\pm$ 1.6	49.4 $\pm$ 0.3	0.0 $\pm$ 0.0	2.1 $\pm$ 0.04

Table 13: Benchmark on CIFAR10 / resnet18, scenario *SUBCLASS* (deer). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.10 $\pm$ 0.01	–	0.38 $\pm$ 0.06	98.0 $\pm$ 0.1	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	97.7 $\pm$ 0.1	95.7 $\pm$ 0.6	100 (ref)
LDA	0.09 $\pm$ 0.02	0.10 $\pm$ 0.01	0.18 $\pm$ 0.03	97.0 $\pm$ 0.5	99.8 $\pm$ 0.1	24 $\pm$ 0.20
LDA-Mix	0.10 $\pm$ 0.03	0.09 $\pm$ 0.01	0.17 $\pm$ 0.02	96.6 $\pm$ 1.2	99.8 $\pm$ 0.1	24 $\pm$ 0.15
LDA-2C	0.10 $\pm$ 0.03	0.10 $\pm$ 0.01	0.17 $\pm$ 0.03	96.5 $\pm$ 1.3	99.7 $\pm$ 0.2	24 $\pm$ 0.10
DIR	0.23 $\pm$ 0.10	0.83 $\pm$ 0.21	0.30 $\pm$ 0.09	96.3 $\pm$ 2.3	100.0 $\pm$ 0.0	22 $\pm$ 0.14
DIR-2C	0.10 $\pm$ 0.02	0.10 $\pm$ 0.01	0.15 $\pm$ 0.02	96.9 $\pm$ 0.8	99.8 $\pm$ 0.1	22 $\pm$ 0.11
SCRUB	0.31 $\pm$ 0.04	0.28 $\pm$ 0.04	0.28 $\pm$ 0.04	95.0 $\pm$ 0.7	97.8 $\pm$ 1.0	22 $\pm$ 0.10
SalUn	0.37 $\pm$ 0.18	0.91 $\pm$ 0.26	0.37 $\pm$ 0.12	88.7 $\pm$ 8.4	97.2 $\pm$ 5.5	17 $\pm$ 0.14
RL+FT	0.22 $\pm$ 0.09	0.79 $\pm$ 0.16	0.31 $\pm$ 0.10	97.1 $\pm$ 0.7	100.0 $\pm$ 0.0	17 $\pm$ 0.13
GA+FT	0.12 $\pm$ 0.01	0.14 $\pm$ 0.01	0.42 $\pm$ 0.05	97.8 $\pm$ 0.1	100.0 $\pm$ 0.0	75 $\pm$ 0.28
FT	0.13 $\pm$ 0.01	0.17 $\pm$ 0.03	0.44 $\pm$ 0.04	97.9 $\pm$ 0.1	100.0 $\pm$ 0.0	75 $\pm$ 0.37
GA	1.9 $\pm$ 0.30	4.8 $\pm$ 0.15	2.6 $\pm$ 0.40	72.4 $\pm$ 1.2	56.3 $\pm$ 0.9	8.3 $\pm$ 0.08

Table 14: Benchmark on CIFAR10 / resnet18, scenario *SUBCLASS* (frog). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.13 $\pm$ 0.01	–	0.50 $\pm$ 0.06	98.0 $\pm$ 0.1	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	97.5 $\pm$ 0.1	94.3 $\pm$ 0.7	100 (ref)
LDA	0.17 $\pm$ 0.11	0.11 $\pm$ 0.01	0.23 $\pm$ 0.03	92.9 $\pm$ 7.2	99.6 $\pm$ 0.2	22 $\pm$ 0.09
LDA-Mix	0.23 $\pm$ 0.19	0.11 $\pm$ 0.02	0.22 $\pm$ 0.03	90.1 $\pm$ 10.5	99.6 $\pm$ 0.4	22 $\pm$ 0.13
LDA-2C	0.18 $\pm$ 0.09	0.12 $\pm$ 0.02	0.24 $\pm$ 0.04	93.3 $\pm$ 5.7	99.6 $\pm$ 0.5	22 $\pm$ 0.21
DIR	0.28 $\pm$ 0.17	0.69 $\pm$ 0.13	0.29 $\pm$ 0.07	91.4 $\pm$ 9.4	100.0 $\pm$ 0.0	20 $\pm$ 0.10
DIR-2C	0.11 $\pm$ 0.01	0.12 $\pm$ 0.01	0.18 $\pm$ 0.01	96.4 $\pm$ 0.4	99.1 $\pm$ 0.5	20 $\pm$ 0.10
SCRUB	0.27 $\pm$ 0.04	0.25 $\pm$ 0.03	0.35 $\pm$ 0.03	95.2 $\pm$ 0.7	97.2 $\pm$ 0.3	20 $\pm$ 0.07
SalUn	0.59 $\pm$ 0.38	0.87 $\pm$ 0.23	0.37 $\pm$ 0.13	81.2 $\pm$ 12.4	98.7 $\pm$ 2.3	16 $\pm$ 0.16
RL+FT	0.29 $\pm$ 0.17	0.74 $\pm$ 0.14	0.31 $\pm$ 0.09	91.2 $\pm$ 10.0	100.0 $\pm$ 0.0	16 $\pm$ 0.06
GA+FT	0.16 $\pm$ 0.01	0.17 $\pm$ 0.01	0.54 $\pm$ 0.06	97.8 $\pm$ 0.1	100.0 $\pm$ 0.0	68 $\pm$ 0.30
FT	0.16 $\pm$ 0.02	0.21 $\pm$ 0.02	0.58 $\pm$ 0.09	97.9 $\pm$ 0.1	100.0 $\pm$ 0.0	68 $\pm$ 0.27
GA	1.4 $\pm$ 0.19	3.6 $\pm$ 0.20	2.8 $\pm$ 0.29	73.9 $\pm$ 0.8	44.6 $\pm$ 4.0	7.6 $\pm$ 0.09

Table 15: Benchmark on CIFAR10 / dinov2-mlp1, scenario *RANDOM* (random N=500). Mean $\pm$ std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_θ	0.01 $\pm$ 0.00	–	0.12 $\pm$ 0.03	96.9 $\pm$ 0.1	100.0 $\pm$ 0.0	–
p_{θ_r}	0.00 $\pm$ 0.00	–	0.00 $\pm$ 0.00	96.9 $\pm$ 0.0	97.2 $\pm$ 0.9	100 (ref)
LDA	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.09 $\pm$ 0.02	96.8 $\pm$ 0.1	100.0 $\pm$ 0.0	1.3 $\pm$ 0.02
LDA-Mix	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.05 $\pm$ 0.02	96.8 $\pm$ 0.1	98.7 $\pm$ 0.4	1.6 $\pm$ 0.70
LDA-2C	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.09 $\pm$ 0.01	96.8 $\pm$ 0.1	99.2 $\pm$ 0.5	1.3 $\pm$ 0.03
DIR	0.03 $\pm$ 0.01	0.04 $\pm$ 0.00	0.10 $\pm$ 0.02	96.4 $\pm$ 0.1	99.5 $\pm$ 0.6	1.2 $\pm$ 0.03
DIR-2C	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.10 $\pm$ 0.03	96.8 $\pm$ 0.1	100.0 $\pm$ 0.0	1.3 $\pm$ 0.03
SCRUB	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	0.11 $\pm$ 0.03	96.8 $\pm$ 0.1	99.4 $\pm$ 0.5	0.66 $\pm$ 0.08
SalUn	0.03 $\pm$ 0.00	0.90 $\pm$ 0.04	0.09 $\pm$ 0.02	96.5 $\pm$ 0.1	100.0 $\pm$ 0.1	0.49 $\pm$ 0.12
RL+FT	0.03 $\pm$ 0.01	0.24 $\pm$ 0.00	0.11 $\pm$ 0.03	96.5 $\pm$ 0.2	100.0 $\pm$ 0.1	0.64 $\pm$ 0.04
GA+FT	0.03 $\pm$ 0.00	0.03 $\pm$ 0.00	0.05 $\pm$ 0.01	96.7 $\pm$ 0.1	99.1 $\pm$ 0.4	19 $\pm$ 0.28
FT	0.03 $\pm$ 0.00	0.03 $\pm$ 0.00	0.08 $\pm$ 0.02	96.7 $\pm$ 0.1	99.7 $\pm$ 0.2	19 $\pm$ 0.40
GA	0.03 $\pm$ 0.00	27 $\pm$ 7.1	0.03 $\pm$ 0.01	96.7 $\pm$ 0.1	97.9 $\pm$ 0.2	0.21 $\pm$ 0.00

Table 16: Benchmark on CIFAR10 / resnet18, scenario *RANDOM* (random N=500). Mean±std over 5 seeds. RTE is normalised by the RTE retrain. **Bold**: top 3 best on KL columns, best on others (closest to p_{θ_r} for Acc_f).

	KL _t (nats)	KL _{last} (nats)	KL _f (nats)	Acc _t (%)	Acc _f (%)	RTE (%)
p_{θ}	0.42 ± 0.01	–	0.95 ± 0.12	89.6 ± 0.1	100.0 ± 0.0	–
p_{θ_r}	0.00 ± 0.00	–	0.00 ± 0.00	89.7 ± 0.1	90.0 ± 1.5	100 (ref)
LDA	0.42 ± 0.01	0.43 ± 0.01	0.94 ± 0.12	89.5 ± 0.1	100.0 ± 0.0	7.0 ± 0.26
LDA-Mix	0.42 ± 0.01	0.42 ± 0.01	0.88 ± 0.11	89.5 ± 0.1	100.0 ± 0.1	7.0 ± 0.31
LDA-2C	0.42 ± 0.01	0.43 ± 0.01	0.94 ± 0.11	89.6 ± 0.2	99.9 ± 0.1	6.9 ± 0.22
DIR	0.41 ± 0.02	0.40 ± 0.02	0.76 ± 0.07	88.9 ± 0.5	90.6 ± 7.6	4.7 ± 0.14
DIR-2C	0.42 ± 0.00	0.42 ± 0.00	0.87 ± 0.12	89.5 ± 0.2	100.0 ± 0.0	4.6 ± 0.15
SCRUB	0.42 ± 0.01	0.46 ± 0.01	0.92 ± 0.12	89.5 ± 0.1	99.6 ± 0.3	1.7 ± 0.17
SalUn	0.44 ± 0.01	0.64 ± 0.07	0.83 ± 0.10	88.9 ± 0.3	96.5 ± 1.3	1.2 ± 0.04
RL+FT	0.43 ± 0.00	0.47 ± 0.03	0.83 ± 0.10	89.1 ± 0.3	97.7 ± 1.5	0.79 ± 0.02
GA+FT	0.52 ± 0.01	0.66 ± 0.02	1.1 ± 0.11	89.4 ± 0.2	98.9 ± 0.2	70 ± 2.2
FT	0.54 ± 0.04	0.67 ± 0.04	1.1 ± 0.16	89.5 ± 0.3	100.0 ± 0.0	70 ± 2.2
GA	0.42 ± 0.01	0.62 ± 0.04	0.91 ± 0.11	89.5 ± 0.2	99.5 ± 0.3	0.48 ± 0.06

Table 17: Recommendations across forgetting scenarios on CIFAR10, architecture resnet18. Each cell reports $\eta_{\max}$ (top, mean±std over 5 seeds) and the percentage of seeds for which $\text{KL}_{\text{net_proxy_before}} < \text{KL}_{\text{net_proxy_after}}$ (bottom; **bold** when 100%). Sub-keys: CLASS=0, SUBCLASS=0, RANDOM=50.

Scenario	LDA	LDA-Mix	LDA-2C	QDA	QDA-Mix
	0.53 ± 0.04	0.16 ± 0.01	0.53 ± 0.04	0.28 ± 0.03	0.28 ± 0.03
<i>Class</i>	100%	100%	100%	100%	100%
	0.43 ± 0.12	0.02 ± 0.01	0.32 ± 0.11	0.24 ± 0.06	0.09 ± 0.01
<i>Subclass</i>	100%	80%	100%	100%	100%
	0.20 ± 0.40	0.00 ± 0.00	0.06 ± 0.02	0.40 ± 0.49	0.05 ± 0.02
<i>Random</i>	20%	80%	100%	40%	100%

Table 18: Recommendations across forgetting scenarios on CIFAR100, architecture dinov2-mlp2. Each cell reports $\eta_{\max}$ (top, mean±std over 5 seeds) and the percentage of seeds for which $\text{KL}_{\text{net_proxy_before}} < \text{KL}_{\text{net_proxy_after}}$ (bottom; **bold** when 100%). Sub-keys: CLASS=0, SUBCLASS=0, RANDOM=50.

Scenario	LDA	LDA-Mix	LDA-2C	QDA	QDA-Mix
	0.14 ± 0.00	0.02 ± 0.00	0.14 ± 0.00	0.13 ± 0.00	0.13 ± 0.00
<i>Class</i>	100%	100%	100%	100%	100%
	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
<i>Subclass</i>	0%	100%	100%	0%	0%
	0.20 ± 0.40	0.00 ± 0.00	0.01 ± 0.00	0.00 ± 0.00	0.01 ± 0.01
<i>Random</i>	20%	100%	100%	0%	80%

Table 19: Recommendations across architectures on CIFAR10, scenario *Class* (sub-key 6). Each cell reports $\eta_{\max}$ (top, mean±std over 5 seeds) and the percentage of seeds for which $\text{KL}_{\text{net_proxy_before}} < \text{KL}_{\text{net_proxy_after}}$ (bottom; **bold** when 100%).

Architecture	LDA	LDA-Mix	LDA-2C	QDA	QDA-Mix
	0.34 ± 0.00	0.07 ± 0.00	0.36 ± 0.00	0.25 ± 0.00	0.25 ± 0.00
Linear	100%	100%	100%	100%	100%
	0.40 ± 0.00	0.09 ± 0.00	0.43 ± 0.00	0.30 ± 0.00	0.30 ± 0.00
MLP-1	100%	100%	100%	100%	100%
	0.41 ± 0.00	0.09 ± 0.00	0.43 ± 0.00	0.30 ± 0.00	0.30 ± 0.00
MLP-2	100%	100%	100%	100%	100%
	0.46 ± 0.02	0.15 ± 0.00	0.47 ± 0.03	0.26 ± 0.03	0.26 ± 0.03
ResNet-18	100%	100%	100%	100%	100%